Supplementary Material: Study Inventory and Quantitative Evidence

Scope and organization. This supplement supports the survey of 114 text-to-SQL studies. Table S1 gives primary assignments, study objectives and the survey's rationales; Table S2 identifies model roles, examined source versions and evidence locations. Table S3 documents token disclosures, and Table S4 lists the BIRD development-set configurations underlying the selection in main-text Table 7. Study IDs and citation keys connect the tables to the manuscript references and source links.

Evidence and assignments. Paper-level extraction records were reconciled with the manuscript, with selective primary-source checks of objectives, model roles and experimental settings. This was not an independent full-paper re-extraction or a multi-coder agreement study. Objectives paraphrase the sources; primary assignments reflect the survey's interpretation of each study's objective and main contribution, rather than an ablation-based rule. The inventory includes proposed systems, empirical evaluations, annotation support and a foundational decoding method. It assigns 66 studies to RC and 48 to EC. Eleven boundary cases, marked B, include an alternative interpretation; these document taxonomy boundaries rather than a classification error rate. Membership alone does not establish accuracy gains or resource savings.

Model and source notation. Table S2 covers models in the selected experimental configurations. G=generator, R=retriever/linker, S=selector, V=verifier, T=teacher, Rt=router and A=other auxiliary role. API and OW are recorded access descriptors, not independently verified weight-access or licensing classifications. U/UNK denotes unresolved or undisclosed information; a size in a model name is not independently inferred. V01 identifies the examined source version. Bibliographic years in Table S1 and source-version dates in Table S2 can differ. All PDF locations use one-based file pages. Study IDs link Tables S3–S4 to the source links in Table S1 and examined versions in Table S2.

Measurement scope. The supplement documents the quantitative summaries, not all experimental measurements in the underlying extraction. Comparisons require alignment of benchmark, configuration and measurement scope. Measured resource use, configured budgets and model-size proxies are distinct, and incomplete source access does not establish non-reporting. Table S3 tallies token disclosures only; model/tool calls, latency and monetary cost are outside this tally.

Data access. The final supplementary PDF containing Tables S1–S4 will be made publicly available upon publication at https://github.com/ChoiGunHee/T2S_RC_and_EC_Supplementary_File.

Table S1: Primary assignments, objectives and survey rationales. B denotes a documented boundary interpretation.

Study and stable key	Primary	Objective and central mechanism	Survey rationale
S001 Rajkumar et al. (Codex evaluation) 2022; source rajkumar2022evaluating	RC1	Evaluate whether Codex and GPT-3 perform competitive text-to-SQL without task-specific fine-tuning. Compare prompt representations and in-domain demonstrations, and analyze Codex errors.	Prompt representations and demonstrations are the variables analyzed; this is an empirical evaluation, not a new pipeline. *Role: Empirical evaluation.*
S002 C3 2023; source dong2023c3	RC1	Improve zero-shot text-to-SQL with ChatGPT. Combine clear prompting, calibration hints and execution-based self-consistency.	Clear prompting and calibration organize zero-shot evidence; execution voting supports the prompting contribution.
S003 DAIL-SQL 2024; source gao2024text	RC1 (B)	Systematically benchmark prompt representations, example selection/organization, fine-tuning and efficiency. DAIL-SQL retrieves examples using question and SQL skeleton similarity and omits demonstration schemas.	Example selection and organization are central; token-efficient serialization is an important alternative EC4 interpretation. *Alternative: EC4.* Demonstration design versus token-efficient context organization.
S004 ACT-SQL 2023; source zhang2023act	RC1	Improve LLM reasoning for text-to-SQL using automatically generated chain-of-thought exemplars. Generate schema-linking CoT automatically and combine static and dynamically selected exemplars; rewrite multi-turn questions.	Automatically constructed reasoning exemplars define the intervention in in-context prompting.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S005 Selective Demonstrations 2023; source chang-fosler-lussier-2023-selective	RC1	Improve cross-domain text-to-SQL with selective demonstrations from out-of-domain and synthetic in-domain sources. ODIS uses SQL-guided similarity and coverage retrieval to combine out-of-domain examples with filtered synthetic in-domain pairs.	Synthetic examples serve demonstration selection and coverage at inference; data synthesis is supporting machinery.
S006 PURPLE 2024; source ren2024purple	RC1	Improve SQL generation by supplying demonstrations with relevant logical operator composition knowledge. PURPLE combines schema pruning, T5 skeleton prediction, four-level automaton retrieval, SQL adaptation and execution voting.	Operator-aware demonstration retrieval is central; trained retrieval and voting support that inference procedure.
S007 Safe-SQL 2025; source lee2025safe	RC1	Improve unsupervised text-to-SQL when suitable training examples are unavailable. Generate ten similar question-SQL-reasoning examples, score semantic/structural/reasoning relevance and filter before inference.	Generated and filtered examples support inference-time prompting when suitable demonstrations are unavailable.
S008 OpenSearch-SQL 2025; source xie2025opensearch	RC1	Improve text-to-SQL using dynamic examples and consistency alignment across modules. OpenSearch-SQL uses preprocessing, extraction, generation, refinement and alignment with self-taught Query-CoT-SQL demonstrations.	Dynamic Query-CoT-SQL examples and alignment anchor the prompting contribution, despite a multi-stage implementation.
S009 De-semanticization / Skeleton Retrieval 2023; source guo2023prompting	RC1	Retrieve better demonstrations by separating question intent from database-specific semantics. Combine de-semanticization, skeleton kNN retrieval, schema filtering and execution-triggered fallback revision.	Removing database-specific semantics improves structural demonstration retrieval; filtering and repair are supporting steps.
S010 CoT Prompting (Tai et al.) 2023; source tai-etal-2023-exploring	RC1	Study how chain-of-thought style prompting can improve multi-step reasoning for text-to-SQL. QDecomp generates all subquestions before SQL; InterCOL adds relevant columns while avoiding intermediate SQL error propagation.	The study evaluates the form of reasoning prompts and subquestion ordering rather than a trained deployment model.
S011 Adapt and Decompose 2023; source arora2023adapt	RC1 (B)	Improve cross-domain and cross-compositional text-to-SQL with reusable prompts. Select a fixed generic prompt covering SQL operators, adapt it to each database offline, and apply three-stage least-to-most inference.	Reusable database-adapted prompts motivate RC1, although its least-to-most workflow also supports RC3. *Alternative: RC3.* Reusable prompt design versus a least-to-most staged workflow.
S012 Divide and Prompt 2023; source liu2023divide	RC1	Improve text-to-SQL reasoning with natural-language decomposition prompts. Divide-and-Prompt provides clause generation, schema linking and generate/refine prompt strategies.	Decomposition is supplied as prompting strategies for clause generation and refinement, rather than a separate orchestration architecture.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S013 PTD-SQL 2024; source luo2024ptd	RC1	Improve reasoning by grouping SQL questions and retrieving targeted demonstrations. PTD-SQL uses a LoRA classifier, automatically constructed drilling banks and semantic/keyword retrieval.	The classifier and retrieval banks select targeted reasoning demonstrations; classifier training alone does not imply EC membership.
S014 CoE-SQL 2024; source zhang2024coe	RC1	Improve multi-turn text-to-SQL through in-context edition chains. CoE-SQL builds SQL edit demonstrations and generates natural-language edits using conversation history.	SQL edition chains organize conversational evidence in in-context examples.
S015 Rose-SQL 2026; source zhou2026rose	RC1	Improve multi-turn SQL generation with small reasoning models without task fine-tuning. Rose-SQL retrieves Role-State transitions and role exemplars with dependency-aware context anchoring.	Role-state exemplars and context anchoring organize multi-turn inference; small model size does not determine the assignment.
S016 RSL-SQL 2024; source cao2024rslsql	RC2	Make schema linking robust to missing relevant elements and prompt noise. RSL-SQL combines bidirectional linking, context augmentation, binary SQL selection and multi-turn correction.	Bidirectional links and context augmentation target missing or distracting schema evidence.
S017 TCSR-SQL 2026; source xu2026tcsr	RC2	Improve text-to-SQL robustness under contextual requirements. TCSR-SQL retrieves static/dynamic schema-content knowledge and revises generated SQL.	Schema and content retrieval supply contextual requirements for subsequent SQL reasoning.
S018 Solid-SQL 2025; source liu2025solid	RC2	Improve SQL robustness to natural-language and schema perturbations. Solid-SQL combines a tuned schema linker, skeleton retrieval and iterative generation.	The central intervention makes schema evidence robust to perturbations; tuning supports that grounding function.
S019 SchemaRAG 2026; source wu2026schemarag	RC2	Improve text-to-SQL using precise schema alignment and retrieved examples. SchemaRAG combines trained SchemaLinker, schema-aware retriever and Pareto-optimal SQL selection.	Trained linkers and retrievers align question-relevant schema evidence; resource effects remain a separate assessment.
S020 CHESS 2024; source talaei2024chess	RC2	Support accurate SQL synthesis across high-budget and constrained deployments. CHESS coordinates information retrieval, schema selection, candidate generation and unit-test selection.	Information retrieval and schema selection ground candidate generation; orchestration and deployment budgets are complementary perspectives.
S021 Before Generation, Align It! 2024; source qu2024before	RC2	Mitigate schema and logic hallucinations through alignment before generation. TA-SQL aligns schema linking and logical synthesis to analogous tasks.	Pre-generation task alignment addresses schema and logic hallucination through grounded evidence.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S022 Extractive Schema Linking 2025; source glass2025extractive	RC2	Improve schema relevance prediction and downstream execution accuracy. ExSL fine-tunes decoder-only LLMs as extractive schema linkers with coarse or role-specific labels.	The learned module directly predicts relevant schema elements; its grounding role, rather than training itself, determines RC2.
S023 DCG-SQL 2025; source lee2025dcg	RC2	Improve demonstration retrieval by contextual question-schema relationships. DCG-SQL builds deep contextual schema-link graphs and trains a graph encoder to retrieve demonstrations.	Graph representations retrieve demonstrations through question-schema relations, making grounded retrieval the central contribution.
S024 Automatic Metadata Extraction 2025; source shkapenyuk2025automatic	RC2	Improve text-to-SQL through automatically extracted database metadata. Profiles and query logs produce field descriptions, schema links and SQL-to-text examples.	Extracted descriptions, links and examples supply database evidence to the generator.
S025 PV-SQL 2026; source tian2026pv	RC2	Improve SQL correctness while limiting inference overhead. PV-SQL probes databases, extracts constraints and applies rule-based verification with iterative repair.	Database probes expose constraints used for verification; the primary analytical focus is evidence-based grounding.
S026 EGREFINE 2026; source wang2026egrefine	RC2	Refine ambiguous schema names while preserving query equivalence. EGREFINE screens names, proposes renamings, verifies with downstream executions and materializes database views.	Renaming and execution-equivalence checks improve schema semantics supplied to downstream generation.
S027 APEX-SQL 2026; source cao2026apex	RC2	Ground SQL reasoning in data via interactive exploration. APEX-SQL combines logical planning, dual-path pruning, parallel profiling and exploration-guided generation.	Interactive profiling tests database hypotheses and grounds subsequent query construction.
S028 SDE-SQL 2026; source xie2026sde	RC2	Ground generation and repair using self-driven SQL probes. SDE-SQL adds exploration before generation and during refinement, with optional tuning of explorer and generator.	Self-driven SQL probes acquire evidence for generation and repair; optional tuning does not change that role.
S029 DIN-SQL 2023; source pourreza2023din	RC3	Improve few-shot text-to-SQL by decomposing generation into simpler tasks. DIN-SQL performs schema linking, complexity classification, decomposed generation and self-correction.	Task decomposition and complexity-dependent stages organize query construction and self-correction.
S030 MAG-SQL 2024; source xie2024mag	RC3	Improve text-to-SQL through iterative subquery construction. MAG-SQL combines soft schema linking, target-condition decomposition, sub-SQL generation and refinement.	Target-condition decomposition and iterative subqueries structure the generation workflow.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S031 MAC-SQL 2025; source wang-etal-2025-mac	RC3	Improve SQL reasoning using cooperating specialized agents. MAC-SQL uses selector, decomposer and refiner agents and releases SQL-Llama trained on agent instructions.	Specialized selector, decomposer and refiner roles define the contribution; agent instruction tuning is supporting machinery.
S032 ReFoRCE 2025; source deng2025reforce	RC3	Handle enterprise SQL tasks with large schemas and ambiguous queries. ReFoRCE compresses database context, self-refines, votes and explores columns for low-confidence cases.	Compression, exploration and consensus are coordinated as an enterprise query-solving workflow.
S033 FlexSQL 2026; source pham2026flexsql	RC3	Enable flexible schema exploration and program execution for enterprise SQL tasks. Diverse plans, SQL/Python execution, code or plan repair, and voting.	Plan construction, SQL/Python execution and repair form a flexible orchestration loop.
S034 MARS-SQL 2026; source yang2026mars	RC3	Train agents to ground schemas, interact with databases and select reliable SQL trajectories. GRPO-trained grounding and generation agents with a supervised trajectory validator.	Training develops agents that ground, generate and validate trajectories; the analyzed contribution is their coordinated workflow.
S035 Decomposition for Enhancing Attention 2024; source xie-etal-2024-decomposition	RC3	Improve SQL generation by decomposing tasks to reduce attention distraction. DEA-SQL filters information, classifies questions, generates SQL, self-corrects and applies active learning.	Information filtering and question classification precede staged generation and correction to manage attention.
S036 Structure-guided LLMs 2025; source zhang2025structure	RC3	Improve SQL generation using explicit query and schema structure. SGU-SQL learns schema links with graph representations and decomposes syntax-tree meta-operations.	Syntax-tree operations structure generation around learned schema relations.
S037 Question Classification / Multi-agent Collaboration 2025; source shao2025enhancing	RC3	Improve generation through query classification and multi-agent reasoning. QCMA-SQL uses a trained schema/difficulty classifier and coordinated SQL agents.	Question classification controls the subsequent agent workflow; its classifier is a supporting component.
S038 Multi-Turn Interactions 2025; source xiong2025multi	RC3	Improve SQL with interactive feedback and context retrieval. Interactive-T2S retrieves schemas, examples and values and iterates SQL with database feedback.	Retrieval and database feedback organize multi-turn interaction during SQL construction.
S039 SQLFixAgent 2025; source cen2025sqlfixagent	RC3	Repair semantic mismatch in SQL generated by fine-tuned models. SQLFixAgent coordinates SQLReviewer, query diversification and SQL refinement around CodeS.	Reviewer, diversification and refinement modules organize a coordinated repair workflow.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S040 MetaSQL 2024; source 10597742	RC4	Improve NL-to-SQL with metadata-controlled candidate generation and ranking. MetaSQL predicts metadata, generates diverse SQL and applies coarse/fine ranking with phrase supervision.	Metadata-controlled generation expands the candidate pool before coarse and fine ranking.
S041 MCS-SQL 2025; source lee-etal-2025-mcs	RC4	Improve SQL by exploring prompt sensitivity and selecting among diverse candidates. MCS-SQL combines multi-prompt schema linking, SQL generation and confidence-filtered multiple-choice selection.	Multiple prompts diversify candidates and multiple-choice judgments select among them.
S042 PET-SQL 2024; source li2024pet	RC4	Improve text-to-SQL with reference-enhanced prompts and cross-model consistency. PET-SQL generates PreSQL, links schema and generates FinSQL with difficulty-aware cross-consistency.	PreSQL and reconstructed schema context support cross-model candidate consistency and selection.
S043 CHASE-SQL 2025; source pourreza2025chasesql	RC4	Improve SQL with diverse reasoning paths and trained pairwise selection. CHASE-SQL combines divide-and-conquer, query-plan reasoning, online synthetic examples, fixing and a preference-trained selector.	Diverse reasoning paths and pairwise selection are central; a trained selector supports the search procedure.
S044 Alpha-SQL 2025; source li2025alphasql	RC4	Enable zero-shot SQL generation without task-specific training. Alpha-SQL uses MCTS, LLM-generated actions, execution feedback and self-consistency rewards.	MCTS actions and execution rewards search alternative SQL paths at inference time.
S045 SQL-o1 2025; source lyu2025sql	RC4	Improve open-model SQL reasoning using guided search and pruning. SQL-o1 uses LoRA cold-start progressive SQL generation and self-reward MCTS search.	Cold-start training supports progressive candidate search; the inference search procedure remains the main analytical focus.
S046 PExA 2026; source parekh2026pexa	RC4 (B)	Improve the latency–accuracy tradeoff of complex SQL agents. PExA plans independent test cases, explores them in parallel and proposes SQL after semantic verification.	Parallel exploration and test cases organize candidate search, with an explicit alternative EC4 latency-budget interpretation. *Alternative: EC4.* Parallel candidate exploration versus latency-aware resource organization.
S047 R3-SQL 2026; source han2026r3	RC4	Improve SQL candidate selection and recover missing correct candidates. R3-SQL groups execution-equivalent queries, combines pointwise/listwise rewards and selectively resamples.	Execution-equivalence grouping, ranking and selective resampling improve candidate coverage and selection.
S048 XiYan-SQL 2026; source liu2026xiyan	RC4	Improve SQL through specialized generators and candidate reorganization. XiYan-SQL combines multi-schema filtering, four tuned generators plus ICL and a tuned selection model.	Multiple specialized generators and a selector define candidate organization; their training does not by itself imply EC membership.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S049 STaR-SQL 2025; source he-etal-2025-star	RC4	Teach SQL reasoning through self-generated rationales and verification. STaR-SQL iteratively trains on correct rationales, resamples difficult cases and uses an outcome reward model for best-of-N.	The stated emphasis is reasoning and additional test-time computation, including outcome-based candidate verification.
S050 CA-SQL 2026; source petullo2026sql	RC4	Allocate SQL search effort according to estimated task complexity. CA-SQL uses schema-subset exploration, evolutionary crossover/mutation and reward-weighted execution consistency.	Complexity estimation controls exploration and recombination within a SQL search procedure.
S051 LEAF-SQL 2026; source tan2026leaf	RC4	Predict robust SQL skeletons with adaptive granularity. LEAF-SQL performs level-wise skeleton search with SkeFor/SkeEva then generates and selects SQL.	Level-wise skeleton exploration and evaluation guide subsequent candidate generation and selection.
S052 Residual Skill Optimization 2026; source zhu2026residual	RC4	Learn complementary prompt skills that recover ensemble residual failures. DIVSKILL-SQL optimizes eight agent skills using GEPA and uses execution deduplication plus pairwise selection.	Complementary prompt skills target residual ensemble failures, making candidate diversity and selection central.
S053 Dynamic-strategy Framework 2026; source su2026robust	RC4	Adapt SQL generation and correction to query complexity. TriSQL combines question-guided schema selection, structure-first generation and complexity-aware refinement/fallback.	Dynamic strategies govern generation, refinement and fallback; the candidate/refinement allocation is the analytical focus.
S054 MCTS-SQL 2026; source yuan2026mctssql	RC4 (B)	Enable lightweight SQL models using feedback-driven search and prefix caching. MCTS-SQL filters schemas, generates initial SQL, searches refinements and caches fixed prefixes.	Feedback-driven MCTS is the organizing mechanism; enabling a lightweight model provides a plausible EC3 interpretation. *Alternative: EC3.* Feedback-driven search versus making a lightweight model viable.
S055 LEVER 2023; source ni2023lever	RC5	Improve language-to-code by learning to verify execution-grounded candidates. LEVER trains a verifier over input/program/execution and combines verifier/generator probabilities with execution marginalization.	A learned verifier scores execution-grounded programs; training supports verification rather than determining the family.
S056 Execution-Consistency Validation 2025; source yang2025automated	RC5 (B)	Validate and fix incorrect text-to-SQL mappings using execution consistency. SQLDriller generates distinguishing databases and executes natural-language queries with LLMs to fix data and predictions.	Execution consistency detects incorrect question-SQL mappings; repaired supervision also supports an EC2 interpretation. *Alternative: EC2.* Validation of query mappings versus acquisition of repaired supervision.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S057 Question Rewriting / Execution-guided Refinement 2024; source mao2024enhancing	RC5	Enhance prompt-based SQL by rewriting questions and using execution feedback. DART-SQL combines question rewriting, content-aware prompts and execution-guided refinement.	Question rewriting and execution feedback diagnose and refine generated SQL.
S058 ToolSQL 2025; source wang2025toolsql	RC5	Repair SQL database mismatches that execution exceptions alone miss. ToolSQL equips a verification/refinement agent with database retrieval and stricter-constraint detection tools.	Database-aware tools expose semantic mismatches for a verification and repair agent.
S059 VET 2026; source wang2026vet	RC5	Make SQL reasoning reliable using verifiable execution traces. VET grounds schema, executes/repairs reasoning steps and checks final SQL against trace and reconstructed question.	Executable reasoning traces provide evidence for stepwise checking and final-query validation.
S060 ErrorLLM 2026; source hong2026errorllm	RC5	Detect and localize SQL errors before targeted refinement. ErrorLLM learns dedicated error tokens with CodeS LoRA and invokes localization/refinement only for detected errors.	Dedicated error detection and localization govern targeted repair; training the detector does not alone imply EC membership.
S061 CSC-SQL 2025; source sheng2025csc	RC5	Improve majority voting by correcting the two most frequent SQL outputs. CSC-SQL combines execution-based voting with a GRPO-trained merge revision model.	A trained merge model corrects frequent candidate outputs, making corrective verification the central contribution.
S062 Dual-State Reasoning 2025; source hao2025text	RC5	Maintain coherent SQL reasoning on large enterprise schemas. DSR-SQL couples adaptive context construction with progressive execution-feedback-guided generation.	Progressive execution feedback and coherent context states guide generation and refinement.
S063 SQL-of-Thought 2025; source chaturvedi2025sqlofthought	RC5	Improve semantic SQL correction beyond execution errors. SQL-of-Thought decomposes schema linking, subproblems, query plans, SQL generation and taxonomy-guided repair across agents.	Taxonomy-guided diagnosis and repair govern the corrective workflow across agents.
S064 MAGIC 2025; source askari2025magic	RC5 (B)	Automate self-correction guidelines rather than manually engineering them. MAGIC coordinates manager, feedback and correction agents over training failures to generate reusable guidelines.	Reusable correction guidelines serve query repair; automatic acquisition of those guidelines also supports EC2. *Alternative: EC2.* Correction procedure versus acquisition and reuse of correction guidelines.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S065 DuBo-SQL 2024; source thorpe2024dubo	EC1 (B)	Improve SQL accuracy while reducing fine-tuning and inference costs. v1 fine-tunes GPT-3.5 with concise prompts; v2 retrieves diverse examples and uses readable schemas and JSON outputs.	The fine-tuned variant addresses adaptation and inference costs; the retrieval-based variant supports an alternative RC1 reading. *Alternative: RC1.* Fine-tuned v1 versus retrieval-based v2; variants have different methodological emphases.
S066 SQLFuse 2024; source zhang2024sqlfuse	EC1	Improve practical SQL generation with schema knowledge and model collaboration. SQLfuse combines schema mining, QLoRA schema linking/generation, execution checks and a hindsight critic.	QLoRA specialization and model collaboration develop reusable schema-linking and generation capability for practical deployment.
S067 JOLT-SQL 2025; source song2025jolt	EC1	Unify schema linking and SQL generation in one efficiently tuned model. JOLT-SQL uses joint discriminative/generative loss and confusion-aware noisy schema sampling.	Joint objectives internalize linking and generation in a shared adapted model, addressing adaptation and task fragmentation.
S068 ROUTE 2025; source qin2025route	EC1	Make small open models robust to schema linking, correction and selection tasks. ROUTE couples multi-task supervised fine-tuning and collaborative prompting with noise filtering.	Multitask adaptation develops robust reusable capabilities in a constrained open-model deployment setting.
S069 Schema on the Inside 2026; source soni2026schema	EC1	Deploy accurate schema-specific SQL with short prompts for a cricket application. Two-phase LoRA training first memorizes the schema then teaches recall from database ID with minimal prompts.	Schema knowledge is internalized through LoRA to support database-ID prompts with less repeated context.
S070 Open-SQL Framework 2024; source chen2024open	EC1	Adapt open language models to Text-to-SQL using prompt design and supervision. Open-SQL combines open schema prompts, CoT supervised tuning, example curation and column truncation.	Curated supervision and prompt design adapt an open model to task and context constraints; openness alone is insufficient.
S071 FinSQL 2024; source 10.1145/3626246.3653375	EC1	Support private financial databases in English and Chinese. FinSQL combines hybrid augmentation, parallel cross-encoder linking, LoRA weight plugins and output calibration.	Adapters and augmentation specialize models to private bilingual financial databases and their domain requirements.
S072 Human-Free Demonstration Fusing 2024; source wang-etal-2024-improving-demonstration	EC2	Reduce dependence on human-labeled demonstrations while improving diversity. FUSED clusters embeddings and iteratively fuses distinct demonstrations into a synthetic pool.	Demonstration fusion expands supervision diversity while reducing dependence on additional manual annotation.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S073 OmniSQL 2025; source DBLP:journals/corr/abs-2503-02240	EC2	Scale diverse SQL supervision using synthetic databases, queries and reasoning. OmniSQL trains QwenCoder models on SynSQL2.5M plus CoT-enhanced Spider/BIRD.	Synthetic databases, SQL pairs and reasoning expand reusable supervision for specialized target models.
S074 Text2SQL-Flow 2025; source cai2025text2sql	EC2	Generate diverse semantically valid SQL data and improve few-shot retrieval. SQL-aware augmentation builds SQLFLOW; fine-tunes generators and a masked question-to-SQL embedding retriever.	SQL-aware augmentation supplies training and retrieval supervision with broader structural coverage.
S075 SING-SQL 2025; source caferoglu2025singsql	EC2	Specialize small SQL models and evaluate them on a specific database. SING-SQL synthesizes schema-covering SQL/question/reasoning pairs and LoRA-tunes in-domain generators.	Schema-covering synthetic data transfers target-database knowledge into an adapted small generator.
S076 Human–LLM Collaborative Annotation 2025; source tian2025text	EC2	Help users create accurate domain-specific Text-to-SQL annotations. SQLsynth integrates grammar SQL generation, LLM question generation/alignment and interactive human correction.	The contribution supplies domain annotations through human-LLM collaboration; downstream SQL accuracy is not required for inclusion. *Role: Annotation support.*
S077 RetrySQL 2026; source rkaczkowska2026retrysql	EC2	Teach small SQL models to correct their own reasoning during generation. Erroneous reasoning steps and BACK tokens are inserted into supervised continuation data.	Failure-containing continuation data teaches reusable self-correction to a constrained student model.
S078 SQuaD-SQL 2026; source wu2026squad	EC2	Transfer SQL ability from large models to a compact student. GPT-4o synthetic pairs are filtered and used for LoRA student adaptation.	Filtered teacher-generated SQL pairs transfer capability into a compact student through efficient adaptation.
S079 DB-Explore 2025; source DBLP:conf/emnlp/MaSLZXPWL25	EC2	Improve small SQL models through database-aware instruction synthesis. Progressive database exploration and filtered synthetic instructions train linker and generator models.	Database exploration creates instructions for adapting constrained linker and generator models.
S080 OpenSQL (Intermediate Supervision) 2026; source hu2026opensql	EC2	Improve the data efficiency of open SQL models using intermediate supervision. Global/local linking, diverse reasoning paths and stepwise tournament selection use intermediate supervision.	Intermediate supervision develops open-model capability with limited training data; inference selection is supporting machinery.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S081 Self-Enhanced Fine-Tuning 2026; source lyu2026integrating	EC2	Combine SQL reasoning and generalization using lightweight supervision. Self-generated principle data, LoRA adaptation and structured CoT with schema selection and correction.	Self-generated principles supply lightweight supervision for adapting task capability.
S082 Weak/Strong LLM Data Synthesis 2024; source yang2024sense	EC2	Use strong and weak synthetic supervision to improve open SQL models. GPT-4 data supports SFT; weak-model preferences support DPO.	Strong-model demonstrations and weak-model preferences supply complementary supervision for an open target model.
S083 CodeS 2024; source li2024codes	EC3	Build specialized open SQL models for benchmark and real-domain deployment. SQL-oriented StarCoder pretraining, database prompt augmentation and bidirectional data augmentation.	Task specialization makes smaller accessible SQL models useful across benchmark and target-domain deployments.
S084 DTS-SQL 2024; source pourreza2024dts	EC3	Improve small SQL models through task decomposition. Schema-linking and SQL-generation stages are separately fine-tuned.	Separately adapted stages make SQL generation feasible for limited-capacity models; decomposition is the enabling mechanism.
S085 DataGPT-SQL-7B 2024; source wu2024datagpt	EC3	Develop compact SQL models using invalid-query feedback. Augmented-data SFT and DPO adapt CodeQwen; a debugger repairs invalid SQL.	Feedback-based adaptation and debugging support a compact SQL generator under capacity constraints.
S086 SLM-SQL 2025; source sheng2025slm	EC3	Improve sub-2B text-to-SQL models using synthetic reasoning and reinforcement learning. SFT and GRPO train generation and merge-revision models for corrective self-consistency.	The objective is useful sub-2B capability; synthetic reasoning and reinforcement learning are means to that deployment goal.
S087 SHARE 2025; source qu2025share	EC3 (B)	Enable precise error localization and efficient correction through small-model assistants. BAM, SAM and LOM assistants recover hidden reasoning and diagnose errors before GPT-4o regenerates SQL.	Small correction assistants address repeated expensive LLM correction; the deployed system can still depend on a large generator. *Alternative: RC5.* Compact correction capability versus query-level verification and repair.
S088 LitE-SQL 2026; source piao2026lite	EC3	Build a compact text-to-SQL system with trained schema retrieval and staged generator tuning. HN-SupCon trains an embedding retriever; SFT and preference refinement train the SQL generator.	Trained retrieval and staged adaptation support a compact generator with limited context and capability.
S089 MSc-SQL 2025; source gorti-etal-2025-msc	EC3 (B)	Develop small accessible SQL models under privacy, accessibility and latency constraints. Schema linking and value retrieval feed three generators; a trained Mistral model selects using candidates and execution results.	Joint candidate critique supports small accessible models under privacy, accessibility and latency constraints. *Alternative: RC4.* Constrained small-model deployment versus multi-candidate critique and selection.

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Study and stable key	Primary	Objective and central mechanism	Survey rationale
S090 Feather-SQL 2025; source pei2025feather	EC3	Adapt multi-stage text-to-SQL prompting to small language models. Schema pruning and compact prompts combine multi-path generation, correction and selection; 1+1 separates chat reasoning from SQL generation.	Compact prompts and divided model roles adapt multi-stage inference to limited-capacity models.
S091 LEARNAT 2025; source liao2025learnat	EC3	Learn reliable SQL decomposition without costly inference-time search. AST-guided synthesis and margin-aware DPO train a single-response SQL model.	Training internalizes decomposition to avoid costly inference-time search in a constrained target model.
S092 CRUSH4SQL 2023; source kothyari2023crush4sql	EC4	Retrieve useful subsets from large database schemas. Hallucinated schema segments guide SGPT collective retrieval before RESDSQL generation.	Collective retrieval selects usable subsets from oversized schemas before generation.
S093 DBCopilot 2023; source wang2025dbcopilot	EC4	Support natural-language queries over massive collections of databases. A T5 generative schema router is trained with reverse synthetic questions and graph-constrained decoding, then prompts GPT-3.5.	The router narrows a massive multi-database search space to a manageable input context.
S094 Gen-SQL 2025; source shi2025gen	EC4	Improve large-schema text-to-SQL retrieval without repeated full-schema LLM scanning. Generate pseudo schema, ground it using embeddings, then iteratively regenerate and debug SQL.	Pseudo-schema retrieval avoids repeated full-schema scanning and controls the information supplied to generation.
S095 LinkAlign 2025; source wang-etal-2025-linkalign	EC4	Scale schema linking to large and ambiguous multi-database environments. Query rewriting, response filtering and schema parsing operate in pipeline or multi-agent modes.	Routing and linking make large multi-database catalogs manageable under practical context constraints.
S096 AutoLink 2026; source wang2026autolink	EC4	Scale schema linking through autonomous exploration rather than full-schema prompts. An agent retrieves, explores and verifies schema before multi-candidate SQL generation and selection.	Autonomous exploration expands only needed schema context instead of prompting with the whole catalog.
S097 SchemaGraphSQL 2026; source safdarian2026schemagraphsql	EC4	Use graph paths for training-free schema linking in large databases. LLM-selected source/destination tables define shortest paths; force-union preserves alternate join paths for generation.	Graph paths retain connected schema subsets while controlling context size.
S098 E-SQL 2024; source caferouglu2024sql	RC2	Improve database alignment through question enrichment instead of explicit schema filtering. Candidate SQL/predicates enrich questions, followed by SQL refinement with execution feedback.	Question enrichment and candidate predicates principally improve schema/data alignment; cost analysis does not make efficiency the primary objective.

Continued

Study and stable key	Primary	Objective and central mechanism	Survey rationale
S099 PICARD 2021; source scholak2021picard	EC4 (B)	Constrain language-model decoding to syntactically and semantically admissible SQL. Incremental parsing with guards rejects inadmissible token predictions during beam search.	Incremental constraints preserve compatibility and avoid very large beams; validity checking also supports RC5. Lower latency is not assumed. *Alternative: RC5.* Compatible constrained inference versus token-level validity checking. *Role: Foundational method.*
S100 TTD-SQL 2025; source sharma2025ttd	EC4	Reduce decoding latency and schema hallucinations with deterministic token completion. Precomputed schema token trees fill unambiguous continuations while LLM decoding handles branches.	Deterministic token completion reduces unnecessary model decoding under schema constraints.
S101 HEXGEN-TEXT2SQL 2025; source chen2025hexgen	EC4	Improve tail latency and throughput of agentic Text-to-SQL serving on heterogeneous GPUs. Adaptive workload-aware dispatching and urgency-based scheduling of multi-stage LLM calls.	Workload-aware scheduling targets serving throughput and tail latency across heterogeneous hardware.
S102 BAP-SQL 2026; source peng2026bapsql	EC4	Train tool-using SQL policies under explicit inference budgets. Budget-aware rewards jointly optimize SQL success and resource consumption under token, tool and execution limits.	Explicit token and interaction budgets define the optimization goal; RL is the means rather than the family criterion.
S103 Knowledge-to-SQL 2024; source hong2024knowledge	EC5	Generate expert knowledge to assist SQL generation without human-provided test-time knowledge. DELLM uses supervised training and database/SQL feedback preference optimization to generate relevant knowledge.	Feedback-based adaptation develops a reusable knowledge provider when expert test-time knowledge is unavailable.
S104 Reasoning-SQL 2025; source pourreza2025reasoning	EC5	Teach small models SQL reasoning through structured partial rewards. GRPO combines execution, schema and structural rewards; trained generator is integrated into CHASE-SQL.	Structured rewards develop reusable SQL capability in a constrained target model; deployment search remains separately accounted for.
S105 FINER-SQL 2026; source hoang2026finer	EC5	Improve compact SQL reasoning with fine-grained rewards and memory-assisted RL. Multi-teacher reasoning distillation followed by GRPO and structural rewards; schema filtering, candidate generation and majority voting.	Distillation and fine-grained rewards develop compact-model reasoning; filtering and voting add deployment costs.
S106 Reward-SQL 2026; source zhang2026reward	EC5	Improve execution-aware stepwise SQL reasoning and selection. CoCTE decomposition; SFT; process-reward-assisted GRPO and best-of-N selection.	Process-reward training develops reusable stepwise SQL capability; best-of-N selection is an additional inference mechanism.

Continued

Study and stable key	Primary	Objective and central mechanism	Survey rationale
S107 SQL-Trail 2026; source hua2026sql	EC5	Train data-efficient SQL agents with adaptive multi-turn interaction. SFT from Claude trajectories followed by GRPO with execution, turn-budget and auxiliary rewards.	Training develops a data-efficient agent with an explicit interaction budget.
S108 Every Step Counts 2026; source dai2026every	EC5	Assign credit to useful intermediate tool steps instead of uniformly rewarding whole trajectories. Rule-based execution process rewards and smoothed step-level advantages for RL.	Step-level credit improves how a fixed target policy acquires tool-use capability from limited reward evidence.
S109 ReEx-SQL 2026; source dai-etal-2026-reex	EC5	Integrate execution feedback into SQL reasoning and reduce sampling overhead. Execution-aware GRPO and tree-pruned self-consistency reuse shared reasoning prefixes.	Execution-aware training develops reusable reasoning while prefix reuse controls additional sampling work.
S110 ReViSQL 2026; source zhu2026revisql	EC5	Improve SQL RL through expert-verified training data and verification-aware rewards. BIRD-Platinum cleaning; CISPO RLVR with VeriEQL and external-knowledge process rewards.	Verified training data and rewards improve capability acquisition under data-quality constraints.
S111 EXPO-SQL 2026; source lee2026expo	EC5	Assign SQL rewards at clause level using execution evidence. Incremental execution and error tracing identify causal clauses for REINFORCE++ updates.	Clause-level execution feedback addresses the credit-assignment bottleneck when adapting a fixed target model.
S112 Arctic-Text2SQL-R1 2026; source yao-etal-2026-arctic	EC5	Identify effective RL training recipes for accurate and generalizable SQL reasoning. Online GRPO from OmniSQL checkpoints with filtered diverse data and simple execution rewards.	The main contribution is a reusable training recipe and specialized checkpoint, with single-model inference as the deployment setting.
S113 ZAS-SQL 2026; source zheng2026zas	EC2 (B)	Transfer failure-derived generation rules to zero-shot settings with limited annotations and context. Offline Map-Reduce rule distillation, schema enrichment, structured prompting and execution-guided early stopping.	Failure-derived rules transfer reusable task knowledge under annotation and context constraints; no weight post-training is required. RC1 remains plausible. *Alternative: RC1.* Failure-derived knowledge transfer versus structured rule-based prompting.
S114 Prompt Design Strategies (Nan et al.) 2023; source nan2023enhancing	RC1	Study demonstration selection and schema representation for in-context SQL generation. Syntax-based similarity/diversity retrieval plus schema augmentation and voting across shot counts.	Similarity-diversity example selection and schema representation define the prompting contribution; grounding and voting are supporting mechanisms.

Table S2: Model metadata and evidence provenance. Models are consolidated across the selected configurations.

Study	Models: role; name; size; access	Examined source and evidence location
S001 Rajkumar et al. (Codex evaluation)	G: davinci-codex; UNK; API G: davinci; UNK; API	V01 = arXiv:2204.00498v1; PDF watermark 15 Mar 2022 PDF pp. 1-4, Sections 1-4; authors Rajkumar, Li, Bahdanau.
S002 C3	G,R: GPT-3.5-Turbo-0301; UNK; API	V01 = arXiv:2307.07306v1, 14 Jul 2023 (PDF watermark) PDF pp. 1,3-7, Sections 1,4,5.1; Dong et al.
S003 DAIL-SQL	G,A: GPT-4; UNK; API A: Graphix; UNK; OW R: all-mpnet-base-v2; UNK; OW	V01 = arXiv:2308.15363v4, 20 Nov 2023 (PDF watermark) PDF pp. 1,5-6,15-16, Sections 1,3.3, Appendix A; Gao et al.
S004 ACT-SQL	G,A: GPT-3.5-turbo; UNK; API R: text2vec-base-chinese; UNK; OW G: GPT-4; UNK; API	V01 = Findings of EMNLP 2023, pp. 3501-3532 PDF pp. 1,3-5, Sections 1,3; Zhang, Cao, Chen, Xu, Yu.
S005 Selective Demonstrations	G,A,T,V: code-davinci-002; UNK; API G: CodeLlama-34B-instruct; 34B; OW	V01 = Findings of EMNLP 2023, pp. 14174-14189 PDF pp. 1,4-6, Sections 1,3-4; Shuaichen Chang and Eric Fosler-Lussier.
S006 PURPLE	G: gpt-3.5-turbo-0613; UNK; API A: T5-3B; 3B; OW R: RESDSQL-style schema ranking classifier (checkpoint UNK); UNK; U G: gpt-4-0613; UNK; API	V01 = arXiv:2403.20014v1, 29 Mar 2024 PDF pp. 1-4, Sections I,III-IV; Ren et al.
S007 Safe-SQL	G,S,T: GPT-4o; UNK; API G,S,T: GPT-4o-mini; UNK; API	V01 = EMNLP 2025 proceedings, pp. 19023-19035 PDF pp. 1,3-4, Sections 1,3; Lee, Baek, Kim, Bae, Lee.
S008 OpenSearch-SQL	G,A,T: GPT-4o-0513; UNK; API R: bge-large-en-v1.5; UNK; OW	V01 = Proc. ACM Manag. Data 3(3), Article 194, June 2025; DOI 10.1145/3725331 PDF pp. 1,4-12, Sections 1,3; Xie, Xu, Zhao, Guo.
S009 De-semanticization / Skeleton Retrieval	G: text-davinci-003; UNK; API R: Pre-trained language model (name UNK); UNK; U	V01 = Examined 12-page manuscript; publication/arXiv version unresolved PDF pp. 1,4-8, Sections 1,3; Guo et al.
S010 CoT Prompting (Tai et al.)	G: code-davinci-002; UNK; API	V01 = EMNLP 2023 proceedings, pp. 5376-5393 PDF pp. 1-4, Sections 1,3; Tai, Chen, Zhang, Deng, Sun.
S011 Adapt and Decompose	G,T: GPT-Turbo-3.5; UNK/UNK (paper claim not established); API G,T: Text-da-Vinci-003; UNK/UNK (paper claim not established); API	V01 = GenBench 2023 workshop proceedings, pp. 25-47 PDF pp. 1,4-6, Sections 1,4; Arora et al.
S012 Divide and Prompt	G,V: GPT-3.5-Turbo; UNK; API	V01 = arXiv:2304.11556v1 [cs.CL] 23 Apr 2023 PDF pp. 1-2 Sections 1-2; Liu and Tan.

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Study	Models: role; name; size; access	Examined source and evidence location
S013 PTD-SQL	G: GPT-4-0613; UNK; API Rt: Llama-2-7b; 7B; OW R: text-embedding-ada-002; UNK; API T: LLM for drilling-bank generation (checkpoint UNK); UNK; U G: GPT-3.5-turbo-0613; UNK; API	V01 = arXiv:2409.14082v1 [cs.CL] 21 Sep 2024 PDF pp. 1-5 Sections 1,3; Luo et al.
S014 CoE-SQL	G: GPT-3.5-turbo-16k; UNK; API	V01 = arXiv:2405.02712v1 [cs.CL] 4 May 2024 PDF pp. 1,3-5 Sections 1,3; Zhang et al.
S015 Rose-SQL	G: Qwen3-8B; 8B; OW G: Qwen3-14B; 14B; OW	V01 = arXiv:2605.03720v1 [cs.CL] 5 May 2026 PDF pp. 1-5 Sections 1-3; Zhou et al.
S016 RSL-SQL	G,R,S,V: GPT-4o; UNK; API G,R,S,V: DeepSeek; UNK; API	V01 = arXiv:2411.00073v2 [cs.CL] 26 Nov 2024 PDF pp. 1-6 Sections I,III; Cao et al.
S017 TCSR-SQL	G,V: Gpt-3.5-turbo-0125; UNK; API R: Dmeta-embedding; UNK; U	V01 = CAAI Transactions on Intelligence Technology, 2025, Examined 15-page publication PDF pp. 1,3-8 Sections 1,3
S018 Solid-SQL	G: Llama3-70b; 70B/UNK; API/OW R: Llama3-8B-Instruct; 8B; OW R: bge-large-en-v1.5; UNK; OW G: GPT-4o-mini; UNK; API G: GPT-4; UNK; API	V01 = COLING 2025 proceedings, pp. 9793-9803, Examined 11-page PDF PDF pp. 1-5 Sections 1-3; Liu et al. Llama3-70b has different access/size descriptions across selected configurations; preserved, not silently normalized.
S019 SchemaRAG	G,T: GPT-4o; UNK; API R: Qwen-7B; 7B; OW R: bge-large-en-v1.5; UNK; OW	V01 = Proc. ACM Manag. Data 4(1), Article82, February2026, Examined 26-page PDF PDF pp. 1,5-14 Sections 1,3; Wu et al.
S020 CHESS	G,S,V: Gemini-1.5-pro; UNK; API R: text-embedding-3-small; UNK; API G: DeepSeek Coder 34B; 34B; OW R: GPT-3.5-turbo; UNK; API V: GPT-4-turbo; UNK; API	V01 = arXiv:2405.16755v3 [cs.LG] 25 Nov 2024 PDF pp. 1-6 Sections 1,3; Talaei et al.
S021 Before Generation, Align It!	G,R: gpt-4-32k; UNK; API	V01 = Findings of ACL2024 pp. 5456-5471, Examined 16-page PDF PDF pp. 1-4 Sections 1,3; Qu et al.
S022 Extractive Schema Linking	G,R: DeepSeek Coder 6.7B; 6.7B; OW	V01 = arXiv:2501.17174v1 [cs.DB] 23 Jan 2025 PDF pp. 1-6 Sections 1,3; Glass et al.

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Study	Models: role; name; size; access	Examined source and evidence location
S023 DCG-SQL	G: Llama 3.1-8B-Instruct; 8B; OW R: RoBERTa-large; approximately 355 million (as reported); OW R: BERT-base initialized bi-encoder; approximately 220 million (as reported); OW T: Llama 3.1-8B; 8B; OW G: GPT-4; UNK; API	V01 = ACL2025 pp. 15397-15412, Examined 16-page PDF PDF pp. 1-5 Sections 1,3; Kim et al.
S024 Automatic Metadata Extraction	G: GPT-4o; UNK; API	V01 = Examined 37-page manuscript; abstract says May 2025 at time of writing; version/publication unresolved PDF pp. 1-5 Sections 1-4; Shkapenyuk, Srivastava, Johnson and Ghane
S025 PV-SQL	G,V: GPT-4o; UNK; API	V01 = Findings of ACL2026 pp. 25827-25845, Examined 19-page PDF PDF pp. 1-5 Sections 1,4; source title PV-SQL
S026 EGREFINE	G,V: Qwen3.5-27B; 27B; OW T: DeepSeek-V4-Pro; UNK; API	V01 = arXiv:2605.00628v1 [cs.DB] 1 May 2026 PDF pp. 1-7 Sections I,IV; Wang et al.
S027 APEX-SQL	G,R: GPT-4o; UNK; API S: Contextual-SQL reward model (checkpoint UNK); UNK; U G,R,S: DeepSeek-R1; UNK; API	V01 = arXiv:2602.16720v2 [cs.DB] 31 May 2026 PDF pp. 1,3-5 Sections 1,3; Cao et al.
S028 SDE-SQL	G,V: Qwen2.5-72B-Instruct; 72B; OW	V01 = ACL2026 pp. 2507-2525, Examined 19-page PDF PDF pp. 1-7 Sections 1,3; Xie et al.
S029 DIN-SQL	G,R,Rt,V: GPT-4; UNK; API	V01 = NeurIPS2023 Examined 10-page main paper; exact revision not stated PDF pp. 1-6 Sections 1,4; Pourreza and Rafiei
S030 MAG-SQL	G,R,Rt,V: GPT-4; UNK; API G,R,Rt,V: GPT-3.5; UNK; API	V01 = arXiv:2408.07930v4 [cs.CL] 7 Nov 2024 PDF pp. 1-6 Sections 1,3; Xie et al.
S031 MAC-SQL	G,R,V: GPT-4-32k; UNK; API G,R,V: SQL-Llama (CodeLlama7B); 7B; OW T: GPT-4; UNK; API	V01 = COLING2025 pp. 538-555, Examined 18-page publication PDF pp. 1-5 Sections 1,3-4; Wang et al.
S032 ReFoRCE	G,R,V: o3; UNK; API G: Arctic-Text2SQL-R1-7B; 7B; OW	V01 = arXiv:2502.00675v5 [cs.CL] 3 Jun 2025 PDF pp. 1-6 Sections 1,3; Deng et al.
S033 FlexSQL	G: gpt-oss-120b; 120B; OW A: gpt-oss-20b; 20B; OW G: Claude Sonnet 4.6; UNK; API	V01 = arXiv:2605.02815v2 [cs.CL] 11 Aug 2026 PDF pp. 1,4-6 Sections 1,3-4; Pham et al.
S034 MARS-SQL	G,R,S: Qwen2.5-Coder-7B-Instruct; 7B; OW T: Gemini-2.5-pro; UNK; API T: GPT-4o; UNK; API	V01 = arXiv:2511.01008v2 [cs.CL] 10 May 2026 PDF pp. 1,4-5 Sections 1,3; Yang et al.

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Study	Models: role; name; size; access	Examined source and evidence location
S035 Decomposition for Enhancing Attention	G,V: gpt-4-0613; UNK; API G,V: gpt-3.5-turbo-0613; UNK; API	V01 = Findings ACL2024 pp. 10796-10816, August11-16,2024; Examined 21-page proceedings PDF PDF pp. 1-5 Sections 1-2
S036 Structure-guided LLMs	G: GPT-4; UNK; API R: SGU graph schema linker (checkpoint UNK); UNK; U G: GPT-4o; UNK; API	V01 = arXiv:2402.13284v4 [cs.DB] 10 Jul 2025 PDF pp. 1-7 Sections 1,3
S037 Question Classification / Multi-agent Collaboration	G,V: DeepSeek-V3; UNK; API Rt: DeBERTa-v3; UNK; OW	V01 = Findings NAACL2025 pp. 4340-4349, April29-May4,2025; Examined 10-page proceedings PDF PDF pp. 1-5 Sections 1-4
S038 Multi-Turn Interactions	G: gpt-4o-2024-05-13; UNK; API R: text-embedding-3-large; UNK; API	V01 = CIKM 2025 proceedings, Examined 11-page PDF PDF pp. 1-5 Sections 1,3
S039 SQLFixAgent	G: CodeS-7B; 7B; OW A,V: GPT-3.5-Turbo; UNK; API	V01 = arXiv:2406.13408v6 [cs.CL] 18 May 2025 PDF pp. 1-5 Sections 1,3
S040 MetaSQL	G: LGESQL; UNK; OW Rt: LGESQL multi-label classifier; UNK; OW S: stsb-mpnet-base-v2; UNK; OW S: RoBERTa-large; UNK; OW G: GPT-4; UNK; API	V01 = ICDE 2024 proceedings pp. 1765-1778, Examined 14-page PDF PDF pp. 1-8 Sections I-III
S041 MCS-SQL	G,S: GPT-4 8K; UNK; API R: text-embedding-ada-002; UNK; API	V01 = arXiv:2405.07467v1 [cs.CL] 13 May 2024 PDF pp. 1-6 Sections 1,3-4
S042 PET-SQL	G: GPT-4 (2023.06.13); UNK; API G: CodeLlama-34B; 34B; OW G: SQLCoder-34B; 34B; OW G: InternLM-70B; 70B; OW G: SenseChat-70B; 70B; API	V01 = arXiv:2403.09732v4 [cs.CL] 2 Jun 2024 PDF pp. 1-7 Sections 1,3
S043 CHASE-SQL	G,T,V: Gemini 1.5 Pro; UNK; API S: Gemini 1.5 Flash; UNK; API	V01 = ICLR 2025 conference version, Examined 31-page PDF PDF pp. 1-7 Sections 1-3
S044 Alpha-SQL	G,V: Qwen2.5-Coder-32B; 32B; OW R: text-embedding-3-large; UNK; API G,V: Qwen2.5-Coder-7B; 7B; OW G,V: Qwen2.5-Coder-14B; 14B; OW	V01 = ICML2025, PMLR267, Examined 21-page proceedings PDF PDF pp. 1-7 Sections 1-5
S045 SQL-o1	G,V: Llama3-8B; 8B; OW G,V: Qwen2.5-7B; 7B; OW	V01 = arXiv:2502.11741v3 [cs.DB] 22 May 2025 PDF pp. 1-7 Sections 1-5
S046 PExA	G,V: GPT-o3; UNK; API G: Submission backend not explicitly identified; UNK; U	V01 = ACL2026 short-paper proceedings, Examined 31-page PDF PDF pp. 1-5 Sections 1-4

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Study	Models: role; name; size; access	Examined source and evidence location
S047 R3-SQL	G: Arctic-Text2SQL-R1-32B; 32B; OW S: R3-POINT-32B (Contextual-RM-32B); 32B; OW S: R3-7B (OmniSQL-7B); 7B; OW Rt: Qwen2.5-Coder-7B-Instruct; 7B; OW	V01 = Findings ACL2026 proceedings, Examined 19-page PDF PDF pp. 1-6 Sections 1-4
S048 XiYan-SQL	G: Qwen2.5-Coder-32B-Instruct; 32B; OW G,T: gpt-4o-0806; UNK; API S: Qwen2.5-Coder-7B-Instruct; 7B; OW T: Qwen-Max; UNK; API	V01 = IEEE TKDE38(4), April2026 pp. 2474-2487, Examined 14-page PDF PDF pp. 1-7 Sections I-IV
S049 STaR-SQL	G: Llama-3.1-8B-Instruct; 8B; OW V: Llama-3.1-8B-Instruct + ORM head; 8B; OW	V01 = arXiv:2502.13550v1 [cs.CL] 19 Feb 2025 PDF pp. 1-6 Sections 1,3-4 Objective evidence: Abstract and Section 1.
S050 CA-SQL	G,Rt,V: GPT-4o-mini; UNK; API	V01 = arXiv:2605.08057v1 [cs.CL] 8 May 2026 PDF pp. 1-8 Sections 1,3-4
S051 LEAF-SQL	G,S,T: GPT-4o; UNK; API G,V: Qwen3-14B; 14B; OW	V01 = arXiv:2605.09295v1 [cs.CL] 10 May 2026 PDF pp. 1-9 Sections I-V
S052 Residual Skill Optimization	G,S,T: Opus-4.6; UNK; API A: Sonnet-4.6; UNK; API	V01 = arXiv:2605.21792v1 [cs.CL] 20 May 2026 PDF pp. 1-8 Sections 1-4
S053 Dynamic-strategy Framework	G: Structure-aware SQL generator (checkpoint UNK); UNK; U V: LLM refiner (checkpoint UNK); UNK; U Rt: BERT-based complexity classifier; UNK; U	V01 = Scientific Reports16:7892 (2026), doi10.1038/s41598-026-39128-9, Examined 27-page PDF PDF pp. 1,6-14 method/training/setup
S054 MCTS-SQL	G,V: Qwen2.5-Coder-Instruct-1.5B; 1.5B; OW	V01 = arXiv:2501.16607v3 [cs.DB] 27 Nov 2025 PDF pp. 1-7 Sections 1,3-4
S055 LEVER	G: code-davinci-002; UNK; API V: T5-base; UNK; OW V: T5-large; UNK; OW	V01 = arXiv:2302.08468v3 [cs.LG] 1 Sep 2023 PDF pp. 1-5 Sections 1-4
S056 Execution-Consistency Validation	G,V: GPT-4; UNK; API G: CodeS-3B; 3B; OW	V01 = Proc.ACM Manag.Data3(3), Article134, June2025, Examined 28-page PDF PDF pp. 1-3,9-16 method/evaluation
S057 Question Rewriting / Execution-guided Refinement	G,V: GPT-3.5-turbo-16k-0613; UNK; API	V01 = Findings ACL2024 proceedings pp. 2009-2024, Examined 16-page PDF PDF pp. 1-6 Sections 1,3-4
S058 ToolSQL	G,V: gpt-4; UNK; API	V01 = KDD2025, August3-7 Toronto, Examined 9-page PDF PDF pp. 1-7 Sections 1,3-5
S059 VET	G,V: DeepSeek-Chat (V3); UNK; API A: bge-m3; UNK; OW G,V: GPT-4o; UNK; API	V01 = Findings ACL2026 proceedings, Examined 17-page PDF PDF pp. 1-8 Sections 1,3-5

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Study	Models: role; name; size; access	Examined source and evidence location
S060 ErrorLLM	G,V: GPT-4o; UNK; API V: CodeS-7B (ErrorLLM); 7B; OW G: OpenSearch-SQL upstream model (UNK); UNK; U	V01 = arXiv:2603.03742v2 [cs.CL] 23 Jun 2026 PDF pp. 1-7 Sections 1,3-4
S061 CSC-SQL	G,V: Qwen2.5-Coder-7B-Instruct; 7B; OW G: XiYanSQL-QwenCoder-32B-2412; 32B; OW G,V: Qwen2.5-Coder-3B-Instruct; 3B; OW	V01 = IJCNLP-AACL2025 proceedings pp. 1473-1496, Examined 24-page PDF PDF pp. 1-4 Sections 1-3
S062 Dual-State Reasoning	G,R,V: DeepSeek-R1-0528; UNK; U G,R,V: DeepSeek-V3-0324; UNK; U G,R,V: DeepSeek-V3.1; UNK; U G,R,V: Qwen3-30B-A3B-Instruct-2507; UNK; U G: Backend UNK (not restated in Table7); UNK; U	V01 = arXiv:2511.21402v1 [cs.CL] 26 Nov 2025 PDF pp. 1-5 Sections 1-4
S063 SQL-of-Thought	G,R,V: Claude3Opus; UNK; API	V01 = Examined 13-page preprint under review; exact arXiv version unresolved PDF pp. 1-7 Sections 1-4
S064 MAGIC	G,V: GPT-4 (snapshot UNK); UNK; API T: LLM for manager/feedback (checkpoint UNK); UNK; U	V01 = Examined 9-page publication pp. 23433-23441; precise proceedings version not explicit in the examined source PDF pp. 1-5 Sections Method/Experimental Setup
S065 DuBo-SQL	G,V: gpt-3.5-turbo-0613; UNK; API G,V: gpt-4-0125-preview; UNK; API R: text-embedding-3-large; UNK; API	V01 = arXiv:2404.12560v1 [cs.CL] 19 Apr 2024 PDF pp. 1-5 Sections 1-5
S066 SQLFuse	G: CodeFuse-DeepSeek-33B; 33B; OW R,S: Llama2-70B; 70B; OW	V01 = arXiv:2407.14568v1 [cs.CL] 19 Jul 2024 PDF pp. 1-9 Sections 1-4
S067 JOLT-SQL	G,R: Qwen2.5-Coder-7B-Instruct; 7B; OW G,R: Qwen2.5-Coder-14B-Instruct; 14B; OW	V01 = EMNLP2025 proceedings pp. 6040-6053; Examined 14-page PDF PDF pp. 1-6 Sections 1-4
S068 ROUTE	G,V: Llama3-8B-Instruct; 8B; OW	V01 = ICLR2025 conference version; Examined 24-page PDF PDF pp. 1-7 Sections 1-4
S069 Schema on the Inside	G: Qwen3-8B; 8B; OW G: Gemini2.0Flash; UNK; API	V01 = Examined 8-page publication pp. 40110-40117 PDF pp. 1-6 Sections 1-6
S070 Open-SQL Framework	G: Code Llama-7B; 7B; OW	V01 = arXiv:2405.06674v1 [cs.CL] 4 May 2024 PDF pp. 1-6 Sections 1-5
S071 FinSQL	G: LLaMA2-13B; 13B; OW R: Roberta-large; UNK; OW G: Baichuan2-13B; 13B; OW R: Chinese-Bert-large; UNK; OW	V01 = SIGMOD-Companion2024, June9-15; Examined 13-page publication PDF pp. 1-10 Sections 1-9
S072 Human-Free Demonstration Fusing	G,T: CodeLlama-34b; 34B; OW R: SGPT-125m; 0.125B; OW	V01 = Examined 15-page publication pp. 1193-1207 PDF pp. 1-4 Sections 1-3

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Study	Models: role; name; size; access	Examined source and evidence location
S073 OmniSQL	G: OmniSQL-7B / Qwen2.5-Coder-7B-Instruct; 7B; OW T: Qwen2.5-72B-Instruct; 72B; OW T: Meta-Llama-3.1-8B-Instruct; 8B; OW T: Meta-Llama-3.1-70B-Instruct; 70B; OW T: DeepSeek-Coder-6.7B-Instruct; 6.7B; OW T: DeepSeek-Coder-33B-Instruct; 33B; OW T: DeepSeek-Coder-V2-Lite-Instruct; UNK; OW T: Qwen2.5-7B-Instruct; 7B; OW T: Qwen2.5-14B-Instruct; 14B; OW T: Qwen2.5-32B-Instruct; 32B; OW T: Qwen2.5-Coder-7B-Instruct; 7B; OW T: Qwen2.5-Coder-14B-Instruct; 14B; OW T: Qwen2.5-Coder-32B-Instruct; 32B; OW	V01 = arXiv:2503.02240v2 [cs.CL] 13 Jul 2025 PDF pp. 1-10 Sections 1-6
S074 Text2SQL-Flow	G: Qwen2.5-Coder-7B-Instruct; 7B; OW G,T: GPT-4o; UNK; API R: Qwen3-Embedding-0.6B; 0.6B; OW	V01 = arXiv:2511.10192v4 [cs.CL] 10 Feb 2026 PDF pp. 1-10 Sections I-IV
S075 SING-SQL	G: Qwen2.5-Coder-3B-Instruct / SingSQL-LM; 3B; OW R,T: Gemini-2.5-Flash; UNK; API	V01 = arXiv:2509.25672v1 [cs.AI] 30 Sep 2025 PDF pp. 1-9 Sections 1-7
S076 Human–LLM Collaborative Annotation	T: GPT-4Turbo; UNK; API T: ChatGPT (GPT-4); UNK; API	V01 = IUI2025 March24-27 publication; Examined 28-page PDF PDF pp. 1-13 Sections 1-7
S077 RetrySQL	G: OpenCoder1.5B; 1.5B; OW T: GPT-4o; UNK; API	V01 = Examined 9-page PDF; version unresolved PDF pp. 1-7 Sections 1-5
S078 SQuaD-SQL	G: Qwen-1.5B (generation/checkpointUNK); 1.5B; OW T: GPT-4o; UNK; API	V01 = arXiv:2607.08161v1 [cs.CL] 9 Jul 2026 PDF pp. 1-6 Sections I-IV
S079 DB-Explore	G,R: Qwen2.5-Coder-7B; 7B; OW T: GPT-4o; UNK; API	V01 = Examined 11-page publication pp. 18969-18979 PDF pp. 1-6 Sections 1-4
S080 OpenSQL (Intermediate Supervision)	G,R,S: Qwen2.5-Coder-7B; 7B; OW R: Qwen2.5-Coder-3B; 3B; OW R: gte-large-en-v1.5; UNK; OW T: DeepSeek-R1; UNK; U	V01 = Examined 15-page PDF; version unresolved PDF pp. 1-9 Sections 1-7
S081 Self-Enhanced Fine-Tuning	G,R,V: Llama-3.1-8B; 8B; OW	V01 = arXiv:2606.15598v1 [cs.AI] 14 Jun 2026 PDF pp. 1-9 Sections 1-5
S082 Weak/Strong LLM Data Synthesis	G: SENSE-13B / CodeLLaMA-13B; 13B; OW T: GPT-4; UNK; API T: DeepSeek-Coder-1.3B; 1.3B; OW	V01 = Examined 12-page publication pp. 7864-7875 PDF pp. 1-6 Sections 1-4

Continued

Study	Models: role; name; size; access	Examined source and evidence location
S083 CodeS	G: CodeS-7B / StarCoder-7B; 7B; OW R: Schema item classifier (checkpoint UNK); UNK; OW T: GPT-3.5; UNK; API	V01 = Proc.ACMManagementofData2(3), Article127, June2024;28pages PDF pp. 1-17 Sections 1-9
S084 DTS-SQL	G,R: DeepSeek7B (exactcheckpointUNK); 7B; OW	V01 = Examined 9-page publication pp. 8212-8220 PDF pp. 1-5 Sections 1-3
S085 DataGPT-SQL-7B	G: CodeQwen1.5-7B-Chat / DataGpt-sql-7B; 7B; OW V: LLaMa/Qwen (debuggercheckpointUNK); UNK; U	V01 = PreprintdatedSeptember25,2024; supplied7pages PDF pp. 1-5 Sections 1-4
S086 SLM-SQL	G,A: Qwen2.5-Coder-1.5B-Instruct; 1.5B; OW A: Qwen2.5-Coder-0.5B-Instruct; 0.5B; OW	V01 = arXiv:2507.22478v1 [cs.CL] 30 Jul 2025 pp. 2-4, Tables 1-3; Appendix B-D pp. 8-9
S087 SHARE	G,T: GPT-4o; UNK; API A: Llama-3.1-8B-Instruct; 8B; OW	V01 = arXiv:2506.00391v1 [cs.CL] 31 May 2025 SHARE: An SLM-based Hierarchical Action CorREction Sections 3-4 pp.3-8; Appendix C pp.14-16 Objective evidence: Abstract and Section 1. Small assistants do not make the selected GPT-4o-assisted pipeline an exclusively small-model system.
S088 LitE-SQL	G: Qwen2.5-Coder-7B; 7B; OW R: Qwen3-Embedding-0.6B; 0.6B; OW	V01 = arXiv:2510.09014v2 [cs.CL] 20 Jan 2026 Sections 3-5 pp. 3-7; Appendices A-B pp. 11-13
S089 MSc-SQL	G,R,S: Mistral-7B-v0.3; 7B; OW G: Llama-3-8B; 8B; OW G: Gemma-2-9B; 9B; OW R: Alibaba-NLP/gte-large-en-v1.5; UNK; OW	V01 = Examined 16-page PDF; version unresolved Sections 3-5 pp. 3-7; Appendix B p. 16 Objective evidence: NAACL 2025 paper 2025.naacl-long.107, Abstract. Gemma 9B versus 8B wording conflict: selected configuration follows implementation and Table 1 (9B).
S090 Feather-SQL	G,R,S,V: Prem-SQL-1.3B; 1.3B; OW R,S,V: Yi-Coder-1.5B; 1.5B; OW	V01 = Examined 18-page PDF; version unresolved Sections 3-4 pp. 3-8; Appendices pp. 12-18
S091 LEARNAT	G: Qwen2.5-Coder-7B; 7B; OW T: GLM-4-Plus; UNK; API	V01 = arXiv:2504.02327v2 [cs.CL] 2 Jul 2026 Sections 3-4 pp. 3-10; Appendix F pp. 23-33
S092 CRUSH4SQL	A: OpenAI DaVinci; UNK; API R: SGPT; UNK; OW G: RESDSQL; UNK; OW	V01 = arXiv:2311.01173v1 [cs.CL] 2 Nov 2023 Sections 3-5 pp. 3-9; appendices pp. 12-13
S093 DBCopilot	G: gpt-3.5-turbo-0125; UNK; API Rt,T: T5-base; UNK; OW	V01 = arXiv:2312.03463v3 [cs.CL] 22 Feb 2025 Sections 3-4 pp. 3-11
S094 Gen-SQL	G: Meta-Llama-3-70B-Instruct; 70B; OW R: bge-large-en-v1.5; UNK; OW G: gpt-4o-2024-08-06; UNK; API	V01 = Examined 14-page PDF; version unresolved Sections 3-4 pp. 2-8; Appendix A-B pp. 12-14

Continued

Study	Models: role; name; size; access	Examined source and evidence location
S095 LinkAlign	G: DeepSeek-R1; UNK; OW R: GLM-4-air; UNK; API R: bge-large-en-v1.5; UNK; OW G: DeepSeek-V3; 671B; OW G: GPT-4; UNK; API	V01 = Examined 15-page PDF; version unresolved Sections 3-4 pp. 3-8; Appendices A-G pp. 11-15
S096 AutoLink	G,R: DeepSeek-V3; UNK; API R: BGE-Large-v1.5; UNK; OW G: Gemini-1.5-Pro; UNK; API G: DeepSeek-R1; UNK; API	V01 = arXiv:2511.17190v1 [cs.CL] 21 Nov 2025 Method pp. 2-4; results pp. 5-7; environment and implementation appendices pp. 11-17
S097 SchemaGraphSQL	G,R: Gemini-2.5-Flash; UNK; API	V01 = Examined 15-page PDF; version unresolved Sections 2-6 pp. 2-8; Appendices A-F pp. 11-15
S098 E-SQL	G: GPT-4o-mini; UNK; API G: GPT-4o; UNK; API G: Qwen2.5 Coder 7B Instruct; 7B; OW	V01 = arXiv:2409.16751v2 [cs.CL] 28 Jan 2025 Sections 3-4 pp. 3-11; prompt appendices pp. 14-20 Primary rationale: direct schema alignment is the central contribution (Abstract and Sections 1, 3).
S099 PICARD	G: T5-3B; 3B; OW	EMNLP 2021, pp. 9895-9901, examined 7-page PDF Sections 2-3 pp. 2-5 The reported time measures decoding only; its pairing with the accuracy configuration remains unresolved.
S100 TTD-SQL	G: CodeLlama-7b-Instruct-hf; 7B; OW	V01 = Examined 12-page PDF; version unresolved Sections 3-6 pp. 3-7; appendices pp. 8-12
S101 HEXGEN-TEXT2SQL	G: Llama3.1-70B; 70B; OW G: Qwen3-30B; 30B; OW	V01 = arXiv:2505.05286v3 [cs.DB] 8 Mar 2026 Sections 5-6 pp. 10-16
S102 BAP-SQL	G: Qwen3.5-4B; 4B; OW G: UNK transfer checkpoint; UNK; U	V01 = arXiv:2608.02876v1 [cs.AI] 3 Aug 2026 Sections 3-7 pp. 2-7; appendices pp. 8-10
S103 Knowledge-to-SQL	G: GPT-4; UNK; API A: LLaMA2-13B / DELLM; 13B; OW	V01 = Examined 12-page PDF; version unresolved Sections 3-4 pp. 4-8; implementation appendix p. 12
S104 Reasoning-SQL	G: Qwen2.5-Coder-14B (all rewards); 14B; OW S: Qwen2.5-Coder-14B; 14B; OW R: Gemini-1.5-Pro-002; UNK; API	V01 = arXiv:2503.23157v2 [cs.LG] 1 Apr 2025 Sections 3-5 pp. 3-9; training and cost appendices pp. 14-29
S105 FINER-SQL	G: Qwen2.5-Coder-3B; 3B; OW R: GRAST-SQL-0.6B; 0.6B; OW T: DeepSeek-R1; UNK; U T: GPT-oss-120B; UNK; U T: Qwen2.5-72B-Instruct; UNK; U T: GPT-4o; UNK; U	V01 = arXiv:2605.03465v1 [cs.DB] 5 May 2026 Sections 3-5 pp. 3-12

Continued

Study	Models: role; name; size; access	Examined source and evidence location
S106 Reward-SQL	G: Qwen3-8B; 8B; OW S: Qwen2.5-7B-Coder-Instruct PRM; 7B; OW T: DeepSeek-V3.1; UNK; API	V01 = Examined 27-page PDF; version unresolved Sections 4-7 pp. 8-23
S107 SQL-Trail	G: Qwen2.5-Coder-7B; 7B; OW T: Claude-Sonnet-3.7; UNK; API	V01 = Examined 23-page PDF; version unresolved Sections 3-5 pp. 3-8; appendices pp. 12-23
S108 Every Step Counts	G: Qwen3-4B-Instruct; 4B; OW G: Qwen2.5-7B-Instruct; 7B; OW	V01 = arXiv:2605.04719v2 [cs.CL] 7 May 2026 Sections 3-4 pp. 3-8; appendices pp. 13-19
S109 ReEx-SQL	G: Qwen2.5-Coder-7B-Instruct; 7B; OW	V01 = Examined 24-page PDF; version unresolved Sections 3-4 pp. 3-9; appendices pp. 13-24
S110 ReViSQL	G: Kimi-K2.6; UNK; U G: Qwen3-235B-A22B; 235B; OW	V01 = arXiv:2603.20004v4 [cs.DB] 21 Aug 2026 Sections 3-5 pp. 4-12; appendices pp. 15-19
S111 EXPO-SQL	G: Qwen2.5-Coder-7B; 7B; OW	V01 = Examined 20-page PDF; version unresolved Sections 3-6 pp. 3-9; Appendix B.3, p. 17
S112 Arctic-Text2SQL-R1	G: Arctic-Text2SQL-R1-32B / OmniSQL-32B / Qwen2.5-Coder-32B; 32B; OW G: Arctic-Text2SQL-R1-7B / OmniSQL-7B / Qwen2.5-Coder-7B; 7B; OW G: Arctic-Text2SQL-R1-32B; 32B; OW G: Arctic-Text2SQL-R1-7B; 7B; OW	V01 = Examined 30-page PDF; version unresolved Sections 3-6 pp. 3-9; compute Appendix I, p. 23
S113 ZAS-SQL	G: DeepSeek-V3; UNK; API	V01 = arXiv:2606.08245v1 [cs.CL] 6 Jun 2026 Sections 3-5 pp. 3-8 Primary rationale: reusable rule knowledge under data/context constraints, without weight post-training (Sections 1, 3–4). This version states that its appendices were omitted; resource non-reporting cannot be inferred for those unavailable materials.
S114 Prompt Design Strategies (Nan et al.)	G: code-davinci-002; UNK; API A: Preliminary SQL predictor: UNK; UNK; U G,A: gpt-3.5-turbo; UNK; API	Findings of EMNLP 2023, pp. 14935-14956, examined 22-page PDF Sections 2-4 pp. 2-8; algorithms and result appendices pp. 13-22 Codex is the code-davinci-002 alias used in this source.

Supplementary evidence for the quantitative summaries

The underlying extraction contains 114 study records, 608 configuration records and 1182 measurement records. Tables S3 and S4 identify the study and configuration populations used in the quantitative summaries.

Token disclosure inventory

Table S3 lists 43 studies with token-related evidence. The absolute-quantity tally comprises 31 explicit-value studies (19 RC, 12 EC) and six additional studies with approximate values, ranges, or bounds (5 RC, 1 EC), giving 37 studies (24 RC, 13 EC). The other six comprise two relative-only, one formula-only, and three graph-only studies. In the predefined 50-study representative subset, 16 studies report absolute quantities (13 explicit and three approximate; 9 RC, 7 EC). A superscript asterisk identifies membership in that subset, including members outside the absolute-quantity tally.

Explicit values cover prompt/context lengths, outputs, individual components, dataset totals or offline generation; the measurement scope is specified for each study. S074 reports offline data generation only. Configured token limits, token throughput, per-token prices and dataset-length statistics alone are excluded from the absolute-usage tally. Relative, formula-only and unlabelled graphical evidence is listed in separate classes. K denotes one thousand and M one million.

Table S3: Study-level token disclosure inventory. Explicit and Approx./range/bound rows form the 37-study absolute-quantity tally; the remaining classes are listed separately. An asterisk marks representative-subset membership.

Study	Paper (year)	Family	Disclosure	Reported token quantity / evidence	Measurement scope and interpretation	PDF location
S002	C3 (2023)	RC1	Approx./ range/bound	Approximately 1,000/query	Inference; per query. Input/output coverage is unspecified; this is the C3 paper's estimate, not a subsequent paper's remeasurement.	1
S003	DAIL-SQL (2024)*	RC1	Graph only	Token-count versus performance plots	Inference; token usage versus accuracy plots. No precise numeric labels for the proposed DAIL-SQL settings; axis ticks and third-party remeasurements are not substituted.	12, Fig.7; 21, Fig.9
S006	PURPLE (2024)*	RC1	Explicit	3.86K/query; 0.48K–3.86K across settings	Inference; Spider dev; ChatGPT; default len=3072, num=30. These token values use ChatGPT, whereas the separately reported best accuracy uses GPT-4.	11, Fig.11
S008	OpenSearch-SQL (2025)	RC1	Approx./ range/bound	9,000–25,000 for the pipeline	Inference; pipeline estimate. Author-estimated range, not a statistical confidence interval; module ranges are not summed to derive another total.	20, Table 6
S013	PTD-SQL (2024)	RC1	Explicit	Mean 1,855/query	Inference; Spider dev; GPT-3.5. Input/output coverage is unspecified.	18, Appendix D, Table 23
S015	Rose-SQL (2026)	RC1	Explicit	SParC: input 17,305/output 1,581; CoSQL: input 14,862/output 1,436	Inference; SQL generator stage; SParC and CoSQL. Generator-stage consumption is distinguished from the entire pipeline.	15, Table 11
S016	RSL-SQL (2024)*	RC2	Explicit	Input 21.90M/output 0.73M	Inference; entire BIRD dev dataset. Dataset totals, not per-query averages.	9, Table V
S017	TCSR-SQL (2026)*	RC2	Explicit	Mean 607/generated SQL	Inference; TCD test; per generated SQL. These values are for TCD, not Spider or BIRD.	10, Table 4

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Table S3 (continued)

Study	Paper (year)	Family	Disclosure	Reported token quantity / evidence	Measurement scope and interpretation	PDF location
S019	SchemaRAG (2026)*	RC2	Explicit	Context size 17,345 tokens/request	Inference context; UniSQL; final +POSG(5) setting. Per-request context size is not cumulative consumption across all calls.	19, Table 4
S020	CHESS (2024)	RC2	Relative only	5× token reduction	Inference; relative reduction factor; absolute consumption unresolved.	1; 11
S025	PV-SQL (2026)	RC2	Explicit	Input 3,805/output 248; input 4,270/output 315	Inference; BIRD dev; two configurations. Separate configurations; their values must not be added.	8, Table 8
S027	APEX-SQL (2026)	RC2	Approx./ range/bound	Approximately 7K	Inference context; verification schema context. Context corresponding to a mean of 383 retained columns; not cumulative pipeline consumption.	9, §4.4.3
S031	MAC-SQL (2025)*	RC3	Approx./ range/bound	BIRD dev: approximately 10M; Spider dev: approximately 6M	Inference; entire development datasets. Dataset totals, not per-query consumption.	7, §5.4
S032	ReFoRCE (2025)*	RC3	Formula only	23K·v, where v denotes votes	Inference; vote-dependent expression. Per-call versus per-example coverage is ambiguous; the expression is not converted into a fixed total/query.	7, Table 1
S034	MARS-SQL (2026)	RC3	Explicit	Mean 13,325/query	Inference; BIRD dev; full pipeline.	17, Table 11
S035	Decomposition for Enhancing Attention (2024)	RC3	Explicit	Mean 5,611; DEA-SQL-C 3,324	Inference; DEA-SQL and DEA-SQL-C ablation. DEA-SQL-C removes the classification module; no input/output split is provided.	8, Table 6
S038	Multi-Turn Interactions (2025)	RC3	Explicit	Spider: 4,634; BIRD: 4,474 with knowledge/4,715 without; Spider 2.0-lite: 7,424.6/17,531.8 for separate engines	Inference; separate benchmarks and configurations. Benchmark, knowledge and engine settings remain separate.	7, Table 8; 8, Table 14
S039	SQLFixAgent (2025)	RC3	Graph only	Token-consumption bar chart	Inference; token-consumption plot with unlabelled bars.	7, Fig.6
S042	PET-SQL (2024)	RC4	Explicit	Mean prompt length 924.2 → 630.8	Inference input context; before/after PreSQL schema linking. Prompt lengths, not total consumption over multiple calls.	9, §4.4
S045	SQL-o1 (2025)	RC4	Explicit	Mean 9,683	Inference; BIRD dev; Qwen2.5-7B. Input/output coverage is unspecified.	25, Table 11
S046	PExA (2026)*	RC4	Explicit	Mean output: correct 19,631/incorrect 29,312	Inference output; Spider 2.0-Snow; o3; conditioned on correctness. Conditional means, not a pooled mean; generation and reasoning are components of output.	14, Table 10
S048	XiYan-SQL (2026)*	RC4	Approx./ range/bound	Approximately input 6,700/output 205/query	Inference; BIRD dev pipeline; per query. Approximate pipeline values.	11, Table XII
S051	LEAF-SQL (2026)	RC4	Graph only	Token-consumption bar chart	Inference; BIRD; 100 randomly sampled queries. Unlabelled bar heights are not digitized into precise values; workload is the 100-query sample.	11, Fig.11

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Table S3 (continued)

Study	Paper (year)	Family	Disclosure	Reported token quantity / evidence	Measurement scope and interpretation	PDF location
S054	MCTS-SQL (2026)	RC4	Explicit	No cache 2,274 → with cache 864; single-shot 197	Inference; Qwen-Coder-1.5B; efficiency experiment on 500 multi-turn tasks. Cache accounting and token coverage require care; the 500-task efficiency workload remains distinct.	7, Table 4
S058	ToolSQL (2025)*	RC5	Explicit	Mean input 6,886/output 63	Inference; SpiderMismatch. SpiderMismatch is distinct from ordinary Spider dev.	8, Table 4
S059	VET (2026)	RC5	Explicit	Mean Simple 14K/Moderate 16K/Challenging 17K	Inference; BIRD dev; difficulty-specific averages. No pooled mean or input/output split is provided.	12, Table 9
S060	ErrorLLM (2026)*	RC5	Explicit	Detector: input 1,739.4/output 1.45; localization/refinement: input 966.4/output 213.5	Inference; BIRD dev; correction stages. Initial SQL generation is excluded.	11, Table 4
S062	Dual-State Reasoning (2025)	RC5	Explicit	Spider 2.0-Snow: 34.35K; BIRD: 3.67K/3.8K	Inference; schema selection stage. Schema selection only; the two BIRD values refer to different model settings.	8, Table 4
S065	DuBo-SQL (2024)*	EC1	Explicit	v1 total: median 1,686/P95 3,327; v2 total: median 7,970/P95 13,599; v1 output: median 49/P95 80	Inference; BIRD dev; distributional statistics. Medians and P95 values are not means; component quantiles must not be added.	6, §5.2
S069	Schema on the Inside (2026)*	EC1	Approx./ range/bound	Baseline input 17K → <100 in text; 100 in figure	Inference input context; private CriQ cricket dataset. The textual bound and figure label are both retained; benchmark is private CriQ cricket.	1; 6, Fig.5
S074	Text2SQL-Flow (2025)*	EC2	Explicit	Mean 4,909/augmented SQL	Offline training-data generation; per augmented SQL. Offline data-generation consumption only, not service-time inference cost.	11, Table IV
S080	OpenSQL (Intermediate Supervision) (2026)	EC2	Explicit	Input as printed: 9.8+30K; output 3K	Inference; mixed-model input/output reporting. The notation combines costs of different models; the input string is not converted into a single normalized value.	11, Table 5
S081	Self-Enhanced Fine-Tuning (2026)	EC2	Explicit	Mean 2,242	Inference; Spider; resource-evaluation sample of 10 queries. The resource sample is not the same population as full-development-set accuracy.	11, Table 6
S087	SHARE (2025)*	EC3	Explicit	Inference: LLM input 716.30/output 68.32; SLM input 1,731.23/output 132.16. Training-data generation: LLM 1,623.28/66.87; SLM 2,308.75/83.04 (input/output)	Inference correction and separate offline training-data generation; BIRD. Inference values cover SQL correction; offline generation values are separate.	8, Table 6
S091	LEARNAT (2025)	EC3	Explicit	DAIL-setting mean 1.8K; AlphaSQL search 204.5K; offline data generation 145,977K	Inference; DAIL setting and AlphaSQL search; separate offline generation. 1.8K is pooled across benchmarks; 145,977K is separate offline cost.	28, Table 11; 9; 7, Table 1
S094	Gen-SQL (2025)	EC4	Explicit	Spider dev: 1,473; Spider-M dev: 2,348	Inference; separate Spider and Spider-M development datasets. Separate benchmark-specific values.	7, Table 5

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Table S3 (continued)

Study	Paper (year)	Family	Disclosure	Reported token quantity / evidence	Measurement scope and interpretation	PDF location
S095	LinkAlign (2025)*	EC4	Explicit	Pipeline 8,507.4/agent 12,486.5	Inference; Spider 2.0-lite efficiency sample; per-query means. Two execution modes on the efficiency-evaluation sample.	14, Table 8
S096	AutoLink (2026)*	EC4	Explicit	Spider 2.0-lite: 38K/query; BIRD: 8K/query; separate Spider 2.0-lite schema-linking value: 21.2K	Inference; Spider 2.0-lite and BIRD; separate schema-linking report. Total-coverage interpretation requires care; the separate schema-linking value is retained rather than conflated.	6, Tables 2–3; 5
S097	SchemaGraph SQL (2026)	EC4	Explicit	Input 6,666/output 84.2	Inference; BIRD dev; proposed force-union setting. Distinguished from the baseline input 1,892/output 70.2.	5, §5.1
S098	E-SQL (2024)	RC2	Explicit	CSG 12,612/199; QE 16,550/292; SR 7,403/267 (input/output)	Inference; BIRD dev; module-specific values. Module values are not summed into a pipeline total without call-frequency evidence.	11, Table 10
S102	BAP-SQL (2026)	EC4	Relative only	XS -4.5%; S -5.0%; transfer -3.5%	Inference; relative reduction; separate tool-observation lengths. Tool-observation lengths 13.8/30.2 are distinguished from total LLM consumption.	7, Table 3; 6; 5
S107	SQL-Trail (2026)*	EC5	Explicit	Total 3.2K = input 2.7K + output 0.5K	Inference; BIRD dev; 7B final RL setting. Reported values are not multiplied by a sample count without supporting scope evidence.	18, Table 8
S109	ReEx-SQL (2026)	EC5	Explicit	Mean output BIRD 2,563.1/Spider 963.9	Inference output; 7B tree-pruning setting. Output only; neither input nor total token consumption.	8, Table 7

BIRD development-set candidate configurations

Table S4 enumerates all 41 eligible configurations from 27 of the 50 representative studies (19 configurations from 12 RC studies; 22 from 15 EC studies). Eligibility requires BIRD, DEV, execution accuracy (EX), a reported finite absolute percentage, an online full-pipeline result for a proposed full system or proposed variant, a known development split, and a full evaluation subset. Direct numeric results and labelled figure values are eligible; baseline reruns and partial-subset evaluations are excluded. Reward-SQL's two configurations are excluded because the source describes a filtered BIRD development set whose filtering criteria and resulting size are unresolved (S106, p. 14, Sec. 7.1).

The **[T7]** marker identifies the 12 configurations from ten studies displayed in main-text Table 7. Study selection is purposive: the authors selected one eligible study per family for the methodological illustration specified below. The selection reasons are given below.

Selection reasons for Table 7.

Family	Selected study	Reason for selection
RC1	DAIL-SQL (S003)	Illustrates demonstration selection and prompt organization.
RC2	RSL-SQL (S016)	Illustrates bidirectional schema linking and context augmentation for schema grounding.
RC3	MAC-SQL (S031)	Illustrates agents with separate roles and bounded iterative coordination.
RC4	CHASE-SQL (S043)	Illustrates diverse candidate generation followed by pairwise selection.
RC5	ErrorLLM (S060)	Illustrates error localization and selective SQL refinement.
EC1	JOLT-SQL (S067)	Illustrates task-specific adaptation through joint-loss training of schema linking and SQL generation.
EC2	OmniSQL (S073)	Illustrates a model trained on synthetic data, with a same-model comparison of greedy decoding and eight-sample voting.
EC3	SLM-SQL (S086)	Illustrates a compact model with 16 versus 64 generation samples, fixed at eight revision samples, and reported accuracy and completion times.
EC4	AutoLink (S096)	Illustrates retrieval settings, candidate counts, and limits on agent and correction turns.
EC5	FineStep (S108)	Illustrates tool-integrated post-training with step-level execution-based rewards.

Within each selected study, Table 7 uses the first eligible configuration in ascending configuration-ID order. The additional paired settings are OmniSQL's eight-sample voting configuration (`S073-V01-C06`) and SLM-SQL's 16-generation-sample configuration (`S086-V01-C02`); these complement the greedy and 64-generation-sample configurations, respectively. Configuration identifiers are preserved, and every measurement identifier in Table S4 equals its configuration identifier followed by `-R01`. The model column identifies the reported configuration; additional component roles are listed in Table S2. The settings column separates inference, training and other stages where applicable. A setting whose phase or applicability to the reported result is unresolved is identified explicitly.

External-knowledge (EK) status distinguishes Used (8 configurations), Prompt (4), Not used (1) and Unknown (28). Used denotes documented use of provided question-level evidence or hints. Prompt denotes an evidence or hint input in the published method prompt, without separate confirmation of the benchmark-provided evidence setting for the reported result. Not used denotes an explicitly reported evaluation without the provided evidence. Unknown means that the setting is not established in the examined evidence, rather than that evidence was absent or unused. Among the twelve Table 7 configurations, three are Used, four are Prompt and five are Unknown. Source locations distinguish result, setting and EK evidence. Settings describe their reported stages and are not uniform measures of end-to-end cost.

Table S4: All 41 eligible BIRD development-set EX configurations. Bold [T7] marks the 12 rows shown in main-text Table 7. EK denotes external knowledge. EX is reported in percent; source locations use one-based PDF pages.

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
DAIL-SQL (2024) RC1; S003-V01-C35 **[T7]**	GPT-4 DAIL-SQL + GPT-4	54.76	Used	*Inference:* BIRD-specific demonstration, sampling and call budgets are not specified with the leaderboard result.	p. 22 Figure 12 DAIL-SQL + GPT-4 / dev (%) Settings: p. 22, Sec. F.2 and Fig. 12.

Continued on next page

Table S4 (continued)

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
Safe-SQL (2025) RC1; S007-V01-C04	GPT-4o SAFE-SQL / GPT-4o	63.50	Unknown	*Inference:* Generate 10 examples; relevance threshold ≥ 8. *Reported scope:* Three LLM calls are stated in the Spider runtime analysis; the BIRD-specific count is unconfirmed.	p. 6 Table 2 SAFE-SQL / GPT-4o / EX Settings: p. 4, Sec. 3.4; p. 6, Sec. 4.6 (Spider calls); p. 11, Table 7.
Safe-SQL (2025) RC1; S007-V01-C06	GPT-4o-mini SAFE-SQL / GPT-4o-mini	62.10	Unknown	*Inference:* Generate 10 examples; relevance threshold ≥ 8. *Reported scope:* Three LLM calls are stated in the Spider runtime analysis; the BIRD-specific count is unconfirmed.	p. 6 Table 2 SAFE-SQL / GPT-4o-mini / EX Settings: p. 4, Sec. 3.4; p. 6, Sec. 4.6 (Spider calls); p. 11, Table 7.
RSL-SQL (2024) RC2; S016-V01-C01 **[T7]**	GPT-4o RSL-SQL (ours) / GPT-4o	67.21	Prompt	*Inference:* Binary selection between two SQL alternatives; iterative correction with a maximum of N rounds (numerical value unresolved).	p. 7 Table I RSL-SQL (ours) / GPT-4o / EX / BIRD dev EK: pp. 3–4, Sec. III-A and Fig. 2 (optional context/evidence input). Settings: pp. 5–6, Sec. III-E and Alg. 1.
RSL-SQL (2024) RC2; S016-V01-C02	DeepSeek RSL-SQL (ours) / DeepSeek	63.56	Unknown	*Inference:* Binary selection between two SQL alternatives; iterative correction with a maximum of N rounds (numerical value unresolved).	p. 7 Table I RSL-SQL (ours) / DeepSeek / EX / BIRD dev Settings: pp. 5–6, Sec. III-E and Alg. 1.
Solid-SQL (2025) RC2; S018-V01-C07	GPT-4 Solid-SQL + GPT-4	58.90	Unknown	*Inference:* Two-round example retrieval and SQL generation. The BIRD result is not labelled by round.	p. 8 Table 3 Solid-SQL + GPT-4 / EX / BIRD dev Settings: p. 5, Sec. 3.4.2; p. 8, Table 3.
SDE-SQL (2026) RC2; S028-V01-C01	Qwen2.5-72B-Instruct SDE-SQL (SFT)	68.19	Unknown	*Inference:* Zero-shot generation. *Evaluation:* Execution-based evaluation repeated five times to average timeout-related variability.	p. 8 Table 2 SDE-SQL (SFT) / EX / BIRD dev Settings: p. 8, Sec. 4.1.1 (evaluation).
SDE-SQL (2026) RC2; S028-V01-C02	Qwen2.5-72B-Instruct SDE-SQL + Qwen2.5-72B-Instruct	67.67	Unknown	*Inference:* Zero-shot generation. *Evaluation:* Execution-based evaluation repeated five times to average timeout-related variability.	p. 8 Table 2 SDE-SQL + Qwen2.5-72B-Instruct / EX / BIRD dev Settings: p. 8, Sec. 4.1.1 (evaluation).
DIN-SQL (2023) RC3; S029-V01-C03	GPT-4 DIN-SQL + GPT-4 (Ours)	50.72	Used	*Inference:* Greedy decoding, temperature 0; maximum output 350 tokens for self-correction and 600 for other modules. Demonstrations are reduced for BIRD, with the count unspecified.	p. 9 Table 3b DIN-SQL + GPT-4 (Ours) / EX / BIRD dev EK: p. 7, Sec. 5.3 (provided knowledge appended as a hint). Settings: p. 7, Secs. 5.2–5.3.
MAG-SQL (2024) RC3; S030-V01-C01	GPT-4 MAG-SQL + GPT-4	61.08	Unknown	*Inference:* Generation and refinement per subquestion; numerical correction limit unresolved.	p. 7 Table 1 MAG-SQL + GPT-4 / EX / BIRD dev Settings: p. 6, Sec. 3.4.

Continued on next page

Table S4 (continued)

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
MAG-SQL (2024) RC3; S030-V01-C02	GPT-3.5 MAG-SQL + GPT-3.5	57.62	Unknown	*Inference:* Generation and refinement per subquestion; numerical correction limit unresolved.	p. 7 Table 1 MAG-SQL + GPT-3.5 / EX / BIRD dev Settings: p. 6, Sec. 3.4.
MAC-SQL (2025) RC3; S031-V01-C01 **[T7]**	gpt-4-32k MAC-SQL + GPT-4-32k	59.39	Used	*Inference:* GPT-4 context limit 32k; two-shot decomposer; at most 3 error-correction rounds.	p. 6 Table 1 MAC-SQL + GPT-4-32k / EX / BIRD dev EK: p. 3, Alg. 1; pp. 13, 16–18, prompts and Fig. 6. Settings: p. 6, Sec. 5.2; p. 7, Table 4; p. 11, App. A.2–A.3.
MAC-SQL (2025) RC3; S031-V01-C03	SQL-Llama (CodeLlama7B) MAC-SQL + SQL-Llama (CodeLlama7B)	43.94	Unknown	*Inference:* Framework refiner permits at most 3 error-correction rounds. SQL-Llama-specific context and demonstration limits are unresolved; GPT-4 settings do not establish these limits.	p. 6 Table 1 MAC-SQL + SQL-Llama (CodeLlama7B) / EX / BIRD dev Settings: p. 11, App. A.2–A.3 (framework); p. 6, Table 1.
CHASE-SQL (2025) RC4; S043-V01-C01 **[T7]**	Gemini 1.5 Pro CHASE-SQL + Gemini1.5	73.01	Used	*Inference:* Three generator strategies; pairwise selection in both candidate orders. Final candidate count for this result is unresolved.	p. 7 Table 2 CHASE-SQL + Gemini1.5 / EX / BIRD dev EK: p. 21, Sec. A.7.2; p. 27, Sec. A.11 (test-time selector). Settings: pp. 5–7, Secs. 2.4–3 and Alg. 3.
Alpha-SQL (2025) RC4; S044-V01-C01	Qwen2.5-Coder-32B Alpha-SQL / Qwen2.5-Coder-32B	69.70	Unknown	*Inference:* 24 MCTS rollouts; 3 samples per expansion action at temperature 0.8; 5 reward samples at temperature 1.0; at most 10 SQL-revision iterations.	p. 8 Table 2 Alpha-SQL / Qwen2.5-Coder-32B / EX / BIRD dev Settings: p. 7, Sec. 5.2.
Alpha-SQL (2025) RC4; S044-V01-C02	Qwen2.5-Coder-7B Alpha-SQL / Qwen2.5-Coder-7B	66.80	Unknown	*Inference:* 24 MCTS rollouts; 3 samples per expansion action at temperature 0.8; 5 reward samples at temperature 1.0; at most 10 SQL-revision iterations.	p. 8 Table 2 Alpha-SQL / Qwen2.5-Coder-7B / EX / BIRD dev Settings: p. 7, Sec. 5.2.
XiYan-SQL (2026) RC4; S048-V01-C01	gpt-4o-0806 + Qwen2.5-Coder-32B-Instruct XiYan-SQL	73.34	Used	*Inference:* 2 filtered schemas; 5 generators; 10 SQL candidates.	p. 7 Table I XiYan-SQL / EX / BIRD dev Settings: pp. 6–7, Sec. IV-A.
ErrorLLM (2026) RC5; S060-V01-C01 **[T7]**	GPT-4o GPT-4o + ErrorLLM (GPT-4o)	66.23	Unknown	*Inference:* Error-triggered localization and single-pass refinement. *Detector vocabulary:* 32 reserved error tokens, with 12 active error types; these are model-design counts, not consumption limits.	p. 7 Table 1 GPT-4o + ErrorLLM (GPT-4o) / EX / BIRD dev Settings: pp. 3, 6–7, Sec. 3.1, Alg. 2 and implementation details; p. 12, Table 5.

Continued on next page

Table S4 (continued)

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
ErrorLLM (2026) RC5; S060-V01-C03	GPT-4o + OpenSearch-SQL upstream model (UNK) OpenSearch-SQL + ErrorLLM (GPT-4o)	72.29	Unknown	*Inference:* Error-triggered localization and single-pass refinement. *Detector vocabulary:* 32 reserved error tokens, with 12 active error types; these are model-design counts, not consumption limits.	p. 7 Table 1 OpenSearch-SQL + ErrorLLM (GPT-4o) / EX / BIRD dev Settings: pp. 3, 6–7, Sec. 3.1, Alg. 2 and implementation details; p. 12, Table 5.
DuBo-SQL (2024) EC1; S065-V01-C01	gpt-3.5-turbo-0613 Dubo-SQL v1	59.71	Used	*Inference:* Temperature 0; no few-shot examples; maximum output 1000 tokens; optional second correction request; model context limit 4096 tokens.	p. 5 Table 2 Dubo-SQL v1 / EX / BIRD dev Settings: pp. 2–3, Secs. 3–3.1; p. 7, Sec. 7.
DuBo-SQL (2024) EC1; S065-V01-C03	gpt-4-0125-preview Dubo-SQL v2	61.47	Used	*Inference:* Temperature 0; retrieved few-shot examples; optional correction request; model context limit 128000 tokens. The v1 output cap of 1000 tokens is not separately confirmed for v2.	p. 5 Table 2 Dubo-SQL v2 / EX / BIRD dev Settings: pp. 3–4, Sec. 3.2; p. 6, Sec. 5.3; p. 7, Sec. 7.
JOLT-SQL (2025) EC1; S067-V01-C03 **[T7]**	Qwen2.5-Coder-7B-Instruct JOLT-SQL + Qwen2.5-Coder-7B	60.40	Unknown	*Inference:* Greedy decoding; schema-linking threshold 0.05. *Training:* Truncation length 4096 tokens; noisy-schema sampling $\beta = 0.1$ for BIRD (0.2 for Spider).	p. 6 Table 1 JOLT-SQL + Qwen2.5-Coder-7B / EX / BIRD dev Settings: p. 5, Secs. 3.4–3.5; pp. 13–14, App. B.2 and Table 10.
JOLT-SQL (2025) EC1; S067-V01-C06	Qwen2.5-Coder-14B-Instruct JOLT-SQL + Qwen2.5-Coder-14B	64.90	Unknown	*Inference:* Greedy decoding; schema-linking threshold 0.05. *Training:* Truncation length 4096 tokens; noisy-schema sampling $\beta = 0.1$ for BIRD (0.2 for Spider).	p. 6 Table 1 JOLT-SQL + Qwen2.5-Coder-14B / EX / BIRD dev Settings: p. 5, Secs. 3.4–3.5; pp. 13–14, App. B.2 and Table 10.
ROUTE (2025) EC1; S068-V01-C03	Llama3-8B-Instruct Llama3 + ROUTE	57.30	Unknown	*Inference:* Temperature 0.01. *Training:* SFT/MSFT batch size 64 for the 8B model.	p. 7 Table 1 Llama3 + ROUTE / EX / BIRD dev Settings: p. 7, implementation details.
OmniSQL (2025) EC2; S073-V01-C05 **[T7]**	OmniSQL-7B / Qwen2.5-Coder-7B-Instruct OmniSQL-7B / Gre	63.90	Unknown	*Inference:* Greedy decoding; temperature 0.	p. 11 Table 4 OmniSQL-7B / Gre / EX / BIRD dev Settings: p. 10, Sec. 6.1.6.
OmniSQL (2025) EC2; S073-V01-C06 **[T7]**	OmniSQL-7B / Qwen2.5-Coder-7B-Instruct OmniSQL-7B / Maj	66.10	Unknown	*Inference:* 8 candidates at temperature 0.8; majority voting by execution result.	p. 11 Table 4 OmniSQL-7B / Maj / EX / BIRD dev Settings: p. 10, Sec. 6.1.6.
Text2SQL-Flow (2025) EC2; S074-V01-C03	Qwen2.5-Coder-7B-Instruct Seed Data + SQLFLOW / Qwen2.5-Coder-7B-Instruct / Gre	58.00	Unknown	*Inference:* Greedy decoding; temperature 0. *Training:* Full-parameter SFT on Seed Data + SQLFLOW.	p. 9 Table II Seed Data + SQLFLOW / Qwen2.5-Coder-7B-Instruct / Gre / EX / BIRD dev Settings: p. 9, Sec. IV-A.5 and Table II.

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Table S4 (continued)

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
Text2SQL-Flow (2025) EC2; S074-V01-C08	GPT-4o SQLFlow-part+Spider+BIRD SQL / with mask	61.00	Unknown	*Inference:* Five retrieved examples; masked retrieval; greedy decoding at temperature 0. *Retriever training:* Per-device batch size 2; this is not a GPT-4o inference batch.	p. 10 Table III SQLFlow-part+Spider+BIRD SQL / with mask / EX / BIRD dev Settings: pp. 9–10, Sec. IV-A.5; p. 10, Table III.
CodeS (2024) EC3; S083-V01-C02	CodeS-7B / StarCoder-7B SFT CodeS-7B / with provided EK	57.17	Used	*Inference:* Beam search with 4 SQL candidates; select the first executable candidate.	p. 19 Table 6 SFT CodeS-7B / with provided EK / EX / BIRD dev Settings: p. 17, Sec. 9.1.4; EK: p. 15, Sec. 9.1.1 and p. 19, Table 6.
CodeS (2024) EC3; S083-V01-C04	CodeS-7B / StarCoder-7B SFT CodeS-7B / without EK	45.24	Not used	*Inference:* Beam search with 4 SQL candidates; select the first executable candidate.	p. 19 Table 6 SFT CodeS-7B / without EK / EX / BIRD dev Settings: p. 17, Sec. 9.1.4; EK: p. 15, Sec. 9.1.1 and p. 19, Table 6.
DTS-SQL (2024) EC3; S084-V01-C03	DeepSeek7B (checkpoint unresolved) DTS-SQL + DeepSeek7B	55.80	Unknown	*Inference:* Decoding budget unresolved. *Training:* Linker batch size 64, learning rate 10^{-5}; SQL-generator batch size 32, learning rate 5×10^{-5}.	p. 5 Table 5 DTS-SQL + DeepSeek7B / Dev EC (EX in text) / BIRD dev Settings: p. 9, App. A.4.
SLM-SQL (2025) EC3; S086-V01-C01 **[T7]**	Qwen2.5-Coder-1.5B-Instruct SLM-SQL-1.5B SFT+RL+CSC n=64	67.08	Prompt	*Inference:* 64 generation samples; 8 revision samples.	p. 3 Table 2 SLM-SQL-1.5B SFT+RL+CSC n=64 / EX / BIRD DEV EK: pp. 15–16, Secs. F.1–F.2 (generation/revision prompts).
SLM-SQL (2025) EC3; S086-V01-C02 **[T7]**	Qwen2.5-Coder-1.5B-Instruct SLM-SQL-1.5B SFT+RL+CSC n=16	64.84	Prompt	*Inference:* 16 generation samples; 8 revision samples.	p. 9 Table 5 SLM-SQL-1.5B SFT+RL+CSC n=16 / EX / BIRD DEV EK: pp. 15–16, Secs. F.1–F.2 (generation/revision prompts).
SHARE (2025) EC3; S087-V01-C01	GPT-4o GPT-4o + SHARE-8B	64.14	Unknown	*Inference:* One correction iteration; maximum output 1024 tokens.	p. 6 Table 2 GPT-4o + SHARE-8B / EX / BIRD DEV Settings: p. 5, Sec. 4.1; p. 15, App. B.2.
LinkAlign (2025) EC4; S095-V01-C04	GPT-4 LinkAlign* simplified + GPT-4	61.60	Unknown	*Inference:* Simplified LinkAlign*, excluding Steps One and Two; variant-specific budget unresolved. *General framework:* Retrieval top-$k = 5$ and turns adapted to database size; applicability to this simplified variant is unconfirmed.	p. 8 Table 5 LinkAlign* simplified + GPT-4 / EX / Bird DEV Settings: p. 6, Sec. 4.1; pp. 7–8, Sec. 4.2.2 and Table 5.

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Table S4 (continued)

Study / configuration	Model / variant	EX (%)	EK status	Reported configuration settings	PDF location / result row
AutoLink (2026) EC4; S096-V01-C01 **[T7]**	DeepSeek-V3 AutoLink + DeepSeek-V3	66.36	Unknown	*Inference:* Initial retrieval top-$n = 30$ for BIRD (100 for Spider 2.0-lite); subsequent top-$m = 3$; at most 10 agent turns; 5 SQL candidates at temperature 1.0; at most 5 correction turns per candidate.	p. 6 Table 3 AutoLink + DeepSeek-V3 / EX / BIRD DEV Settings: p. 15, More Experiment Details: Experimental Setup.
AutoLink (2026) EC4; S096-V01-C02	Gemini-1.5-pro AutoLink + Gemini-1.5-Pro	68.71	Unknown	*Inference:* Initial retrieval top-$n = 30$ for BIRD (100 for Spider 2.0-lite); subsequent top-$m = 3$; at most 10 agent turns; 5 SQL candidates at temperature 1.0; at most 5 correction turns per candidate.	p. 6 Table 3 AutoLink + Gemini-1.5-Pro / EX / BIRD DEV Settings: p. 15, More Experiment Details: Experimental Setup.
Reasoning-SQL (2025) EC5; S104-V01-C01	Qwen2.5-Coder-14B (all rewards) Reasoning-SQL 14B all rewards + CHASE-SQL	72.29	Unknown	*Inference:* CHASE-SQL pipeline with a Qwen2.5-Coder-14B pairwise selector; candidate and correction budgets are not specified for this result.	p. 8 Table 3 Reasoning-SQL 14B all rewards + CHASE-SQL / EX / BIRD DEV Settings: pp. 7–8, pipeline description and Sec. 5.2; Table 3.
FINER-SQL (2026) EC5; S105-V01-C01	Qwen2.5-Coder-3B FINER-SQL 3B	67.73	Unknown	*Inference:* Top 30 columns; 30 SQL candidates; majority voting. *Training:* 32 rollouts per sample; 20 memory-reward traces.	p. 9 Table II FINER-SQL 3B / EX / BIRD DEV Settings: p. 9, Sec. V-A.
SQL-Trail (2026) EC5; S107-V01-C03	Qwen2.5-Coder-7B SQL-TRAIL 7B majority8	64.20	Unknown	*Inference:* 8 candidates with majority voting; at most 10 turns per question.	p. 6 Table 1 SQL-TRAIL 7B majority8 / EX / BIRD DEV Settings: p. 5, Sec. 4.2; p. 17, Appendix A.8.
Every Step Counts (2026) EC5; S108-V01-C01 **[T7]**	Qwen3-4B-Instruct FineStep 4B SC8	68.51	Prompt	*Inference:* Self-consistency over 8 responses; temperature 1.0. *Training limits:* Prompt 8192 tokens; response 3072; tool response 256; at most 10 assistant turns. Corresponding inference caps are unconfirmed.	p. 6 Table 1 FineStep 4B SC8 / EX / BIRD DEV EK: pp. 18–19, Figs. 8–9 (hint input and BIRD example). Settings: p. 5, Sec. 4 (inference); p. 14, Table 7 (training).